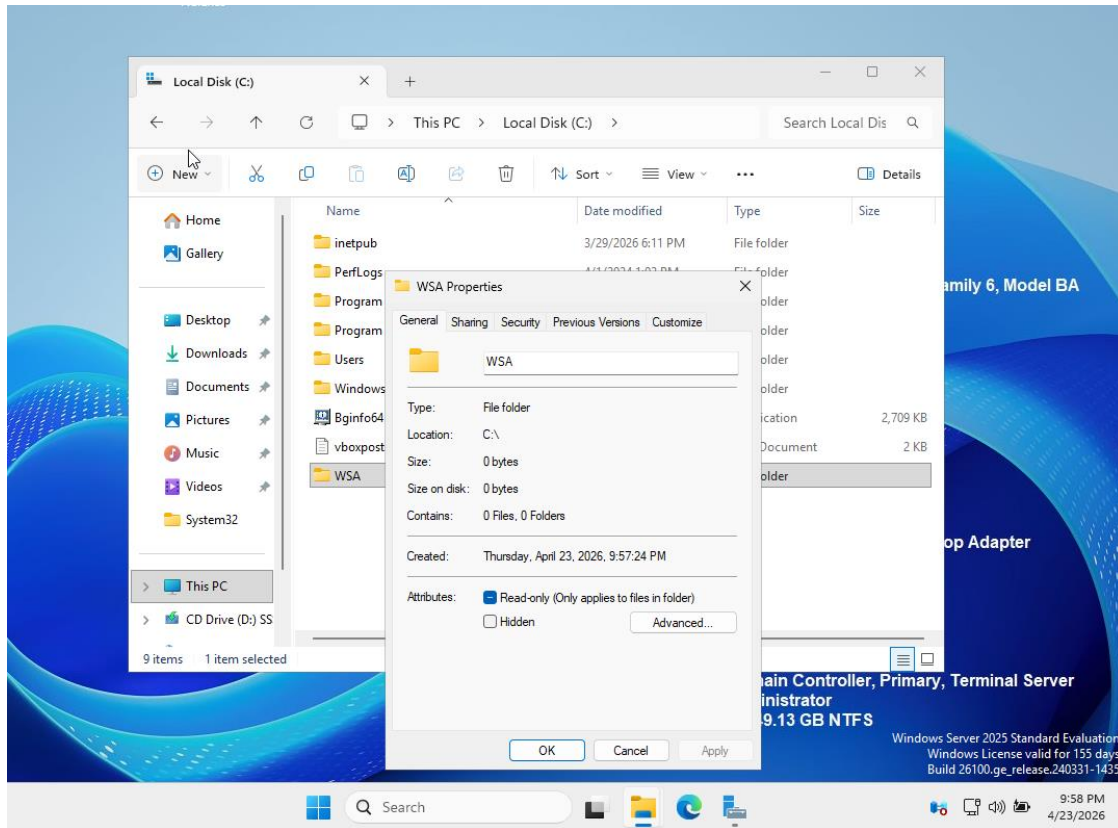
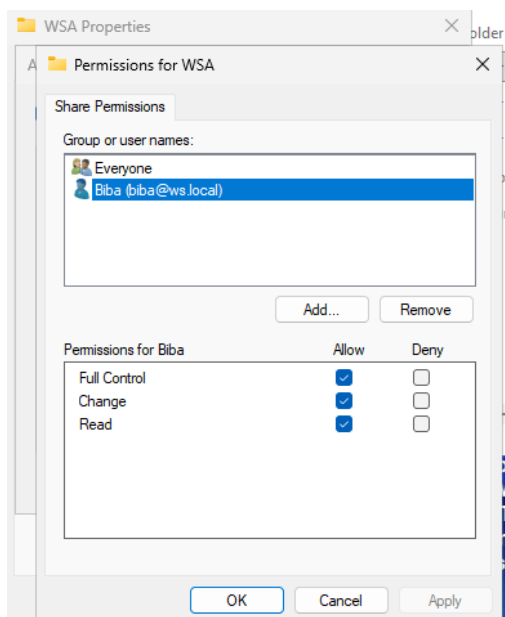

Assignment 5 CS-2424 Bolatbekov Nurzhas

PART – 1 Setting up the network file Server

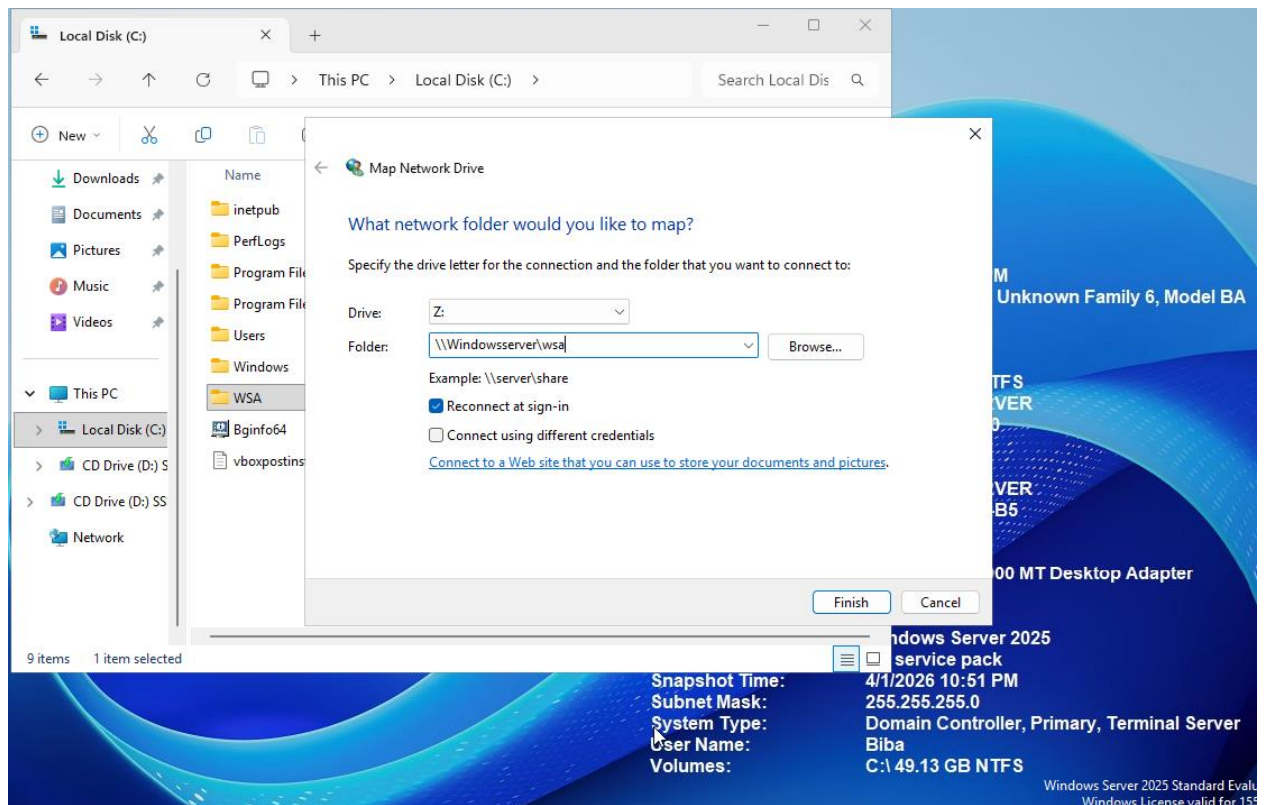
First, I create a folder in Drive C named WSA and open properties:



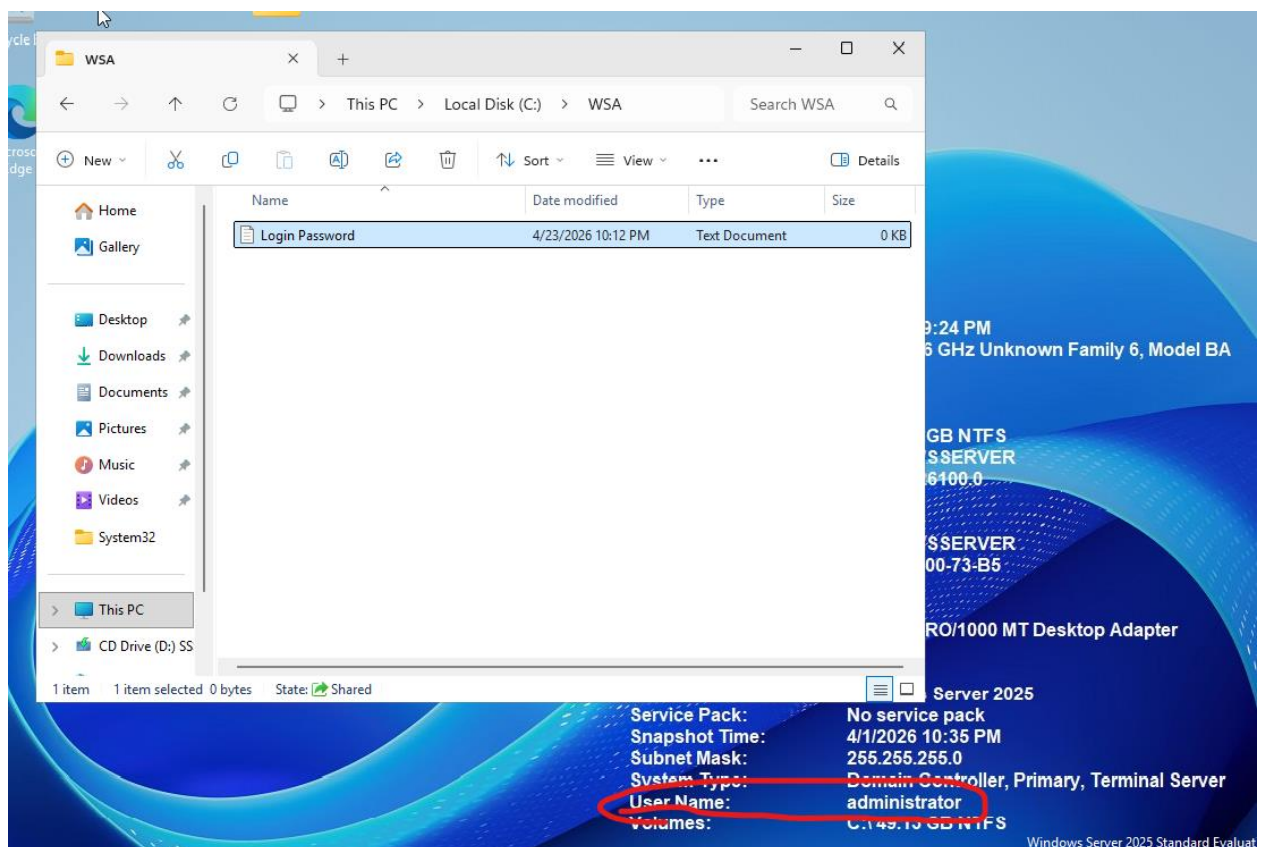
Next I give permissions:



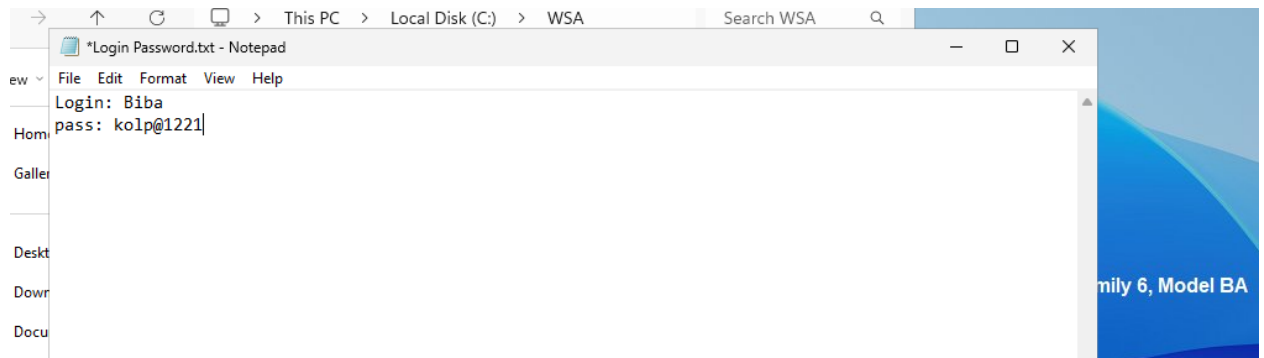
Then I go to Map Network drive and paste network path of WSA:



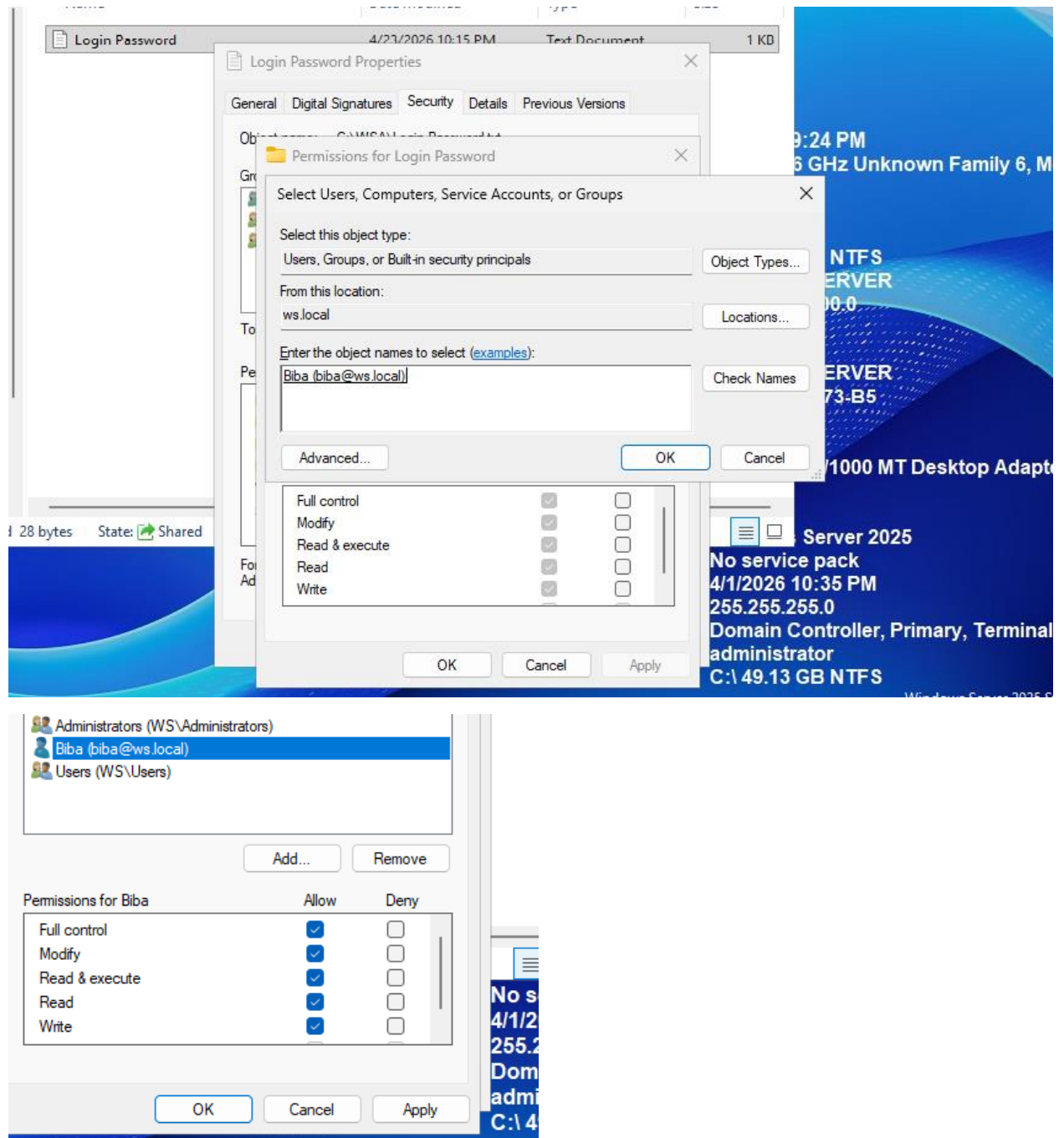
After I change user to admin and create a text file:



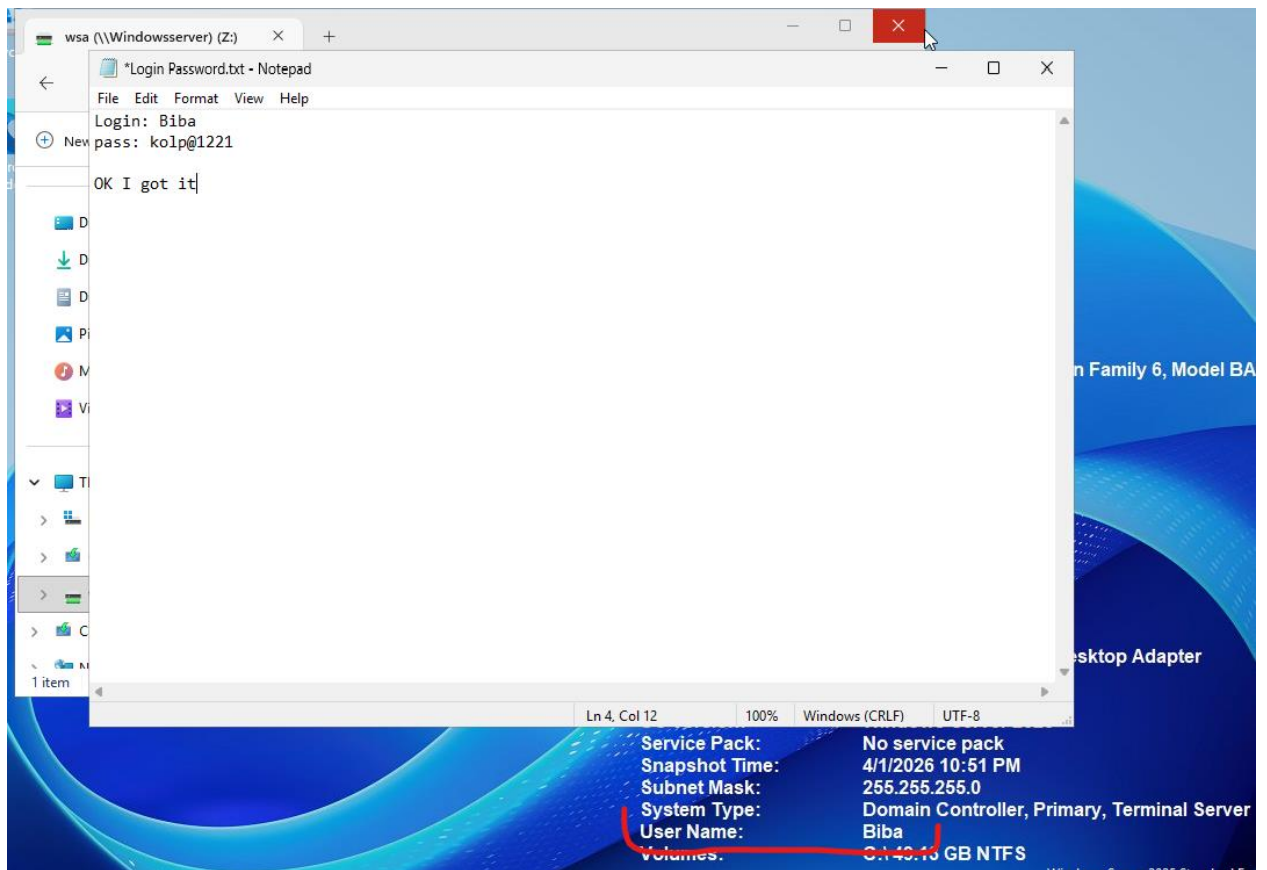
Then I write down info to this text file:



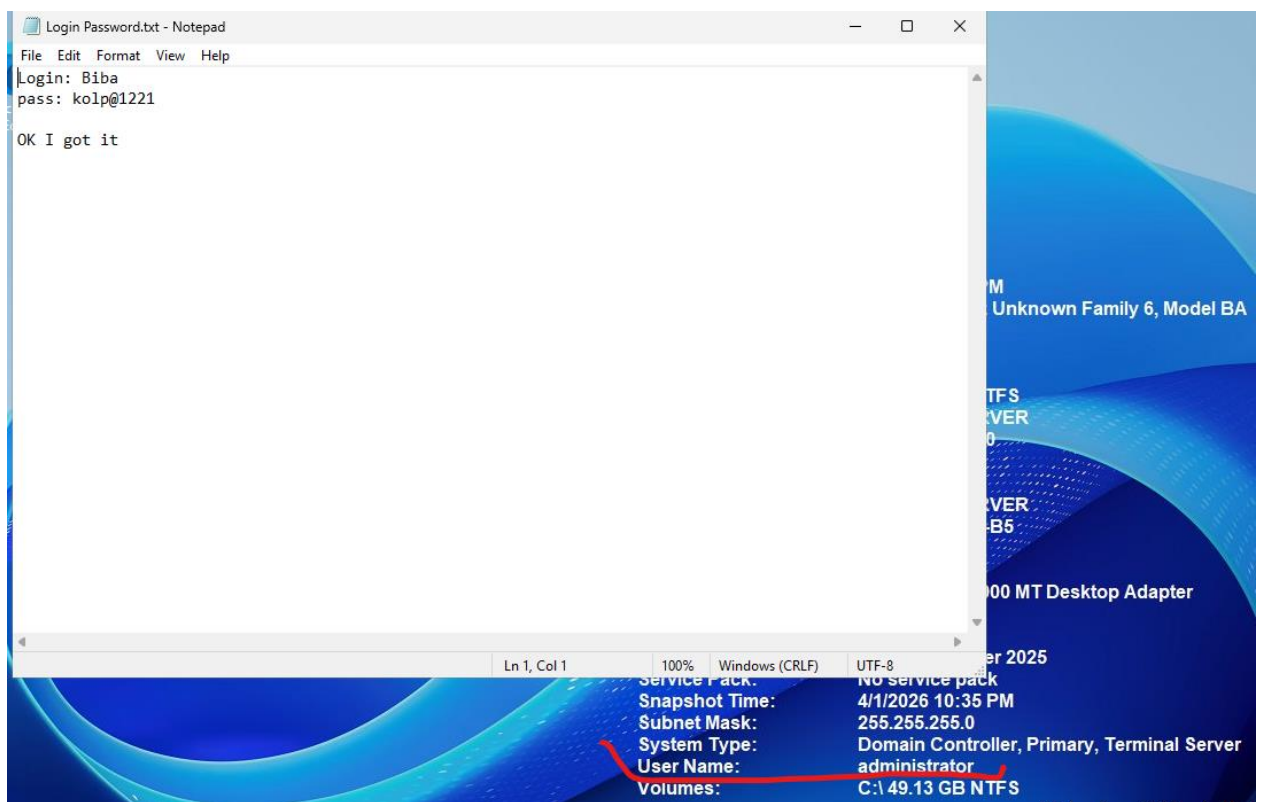
After I change permissions and add user Biba and give full control:



Change user to own user (Biba) and write down that info is correctly showing:

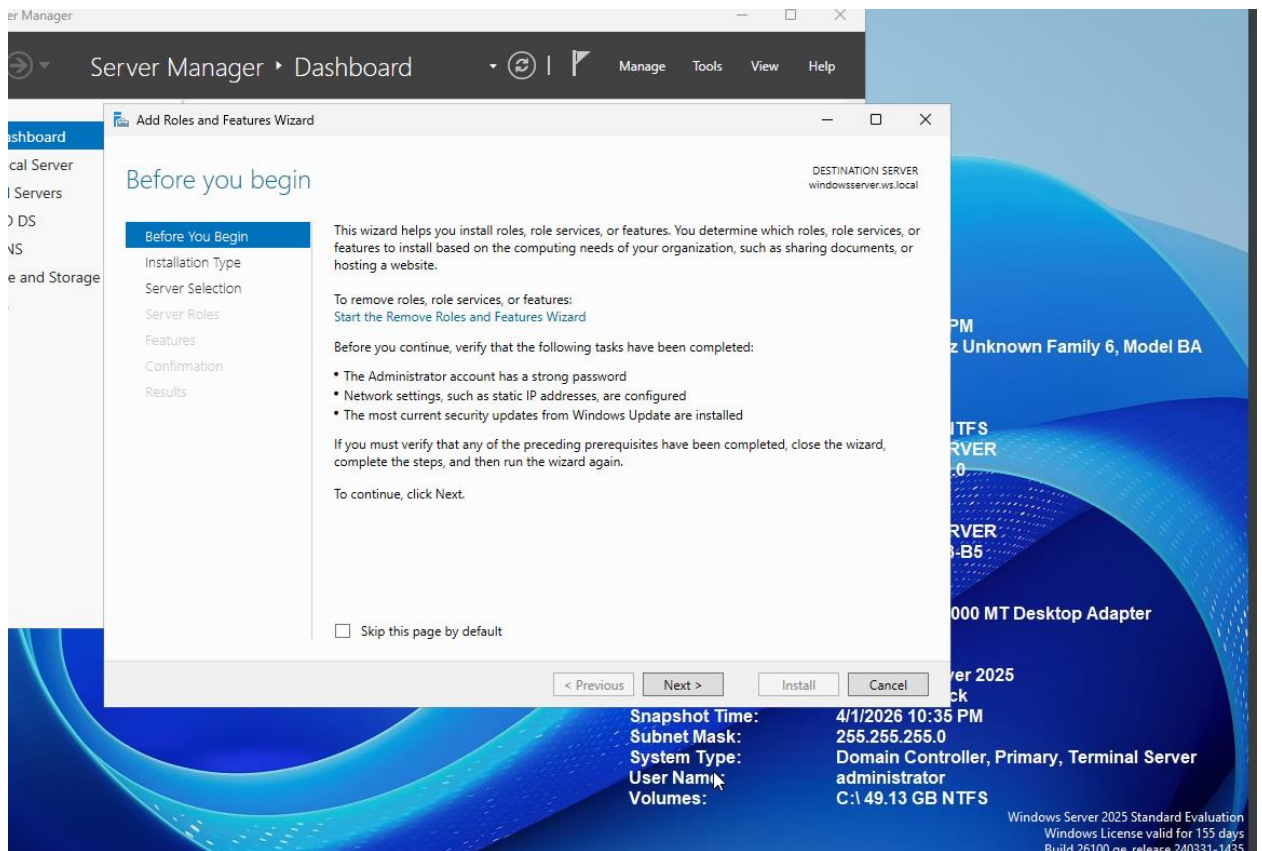


And then Check it from administrator:

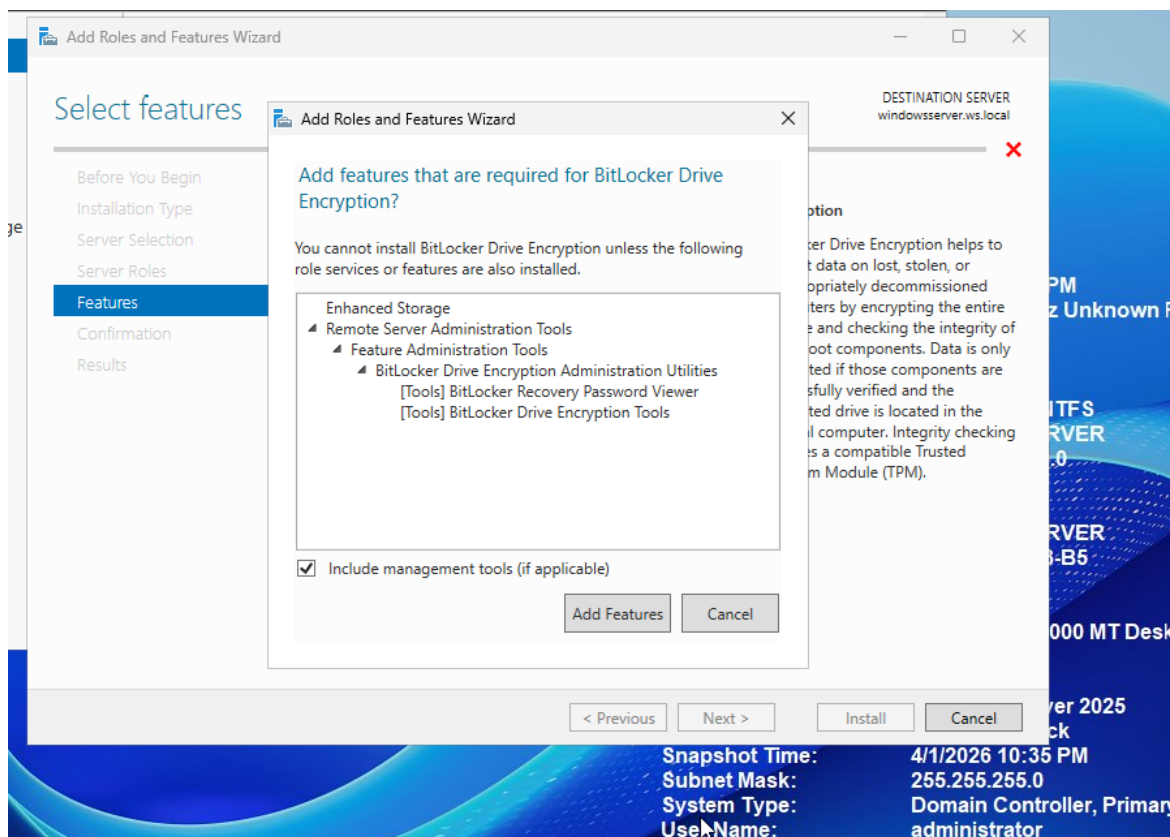


PART – 2 Bitlocker

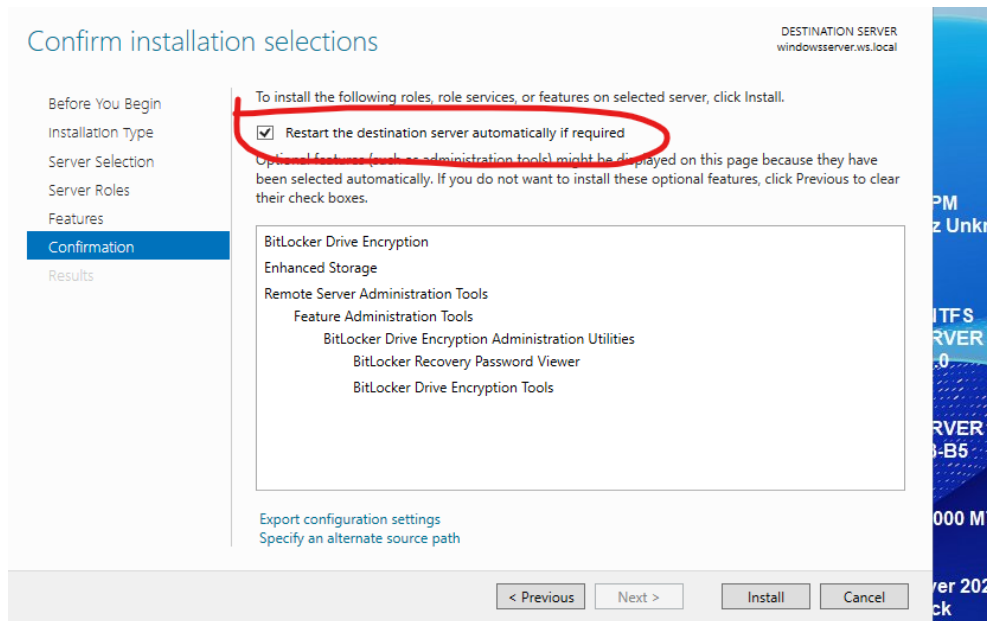
We enter as an admin and open Server Manager and click to Add roles and features



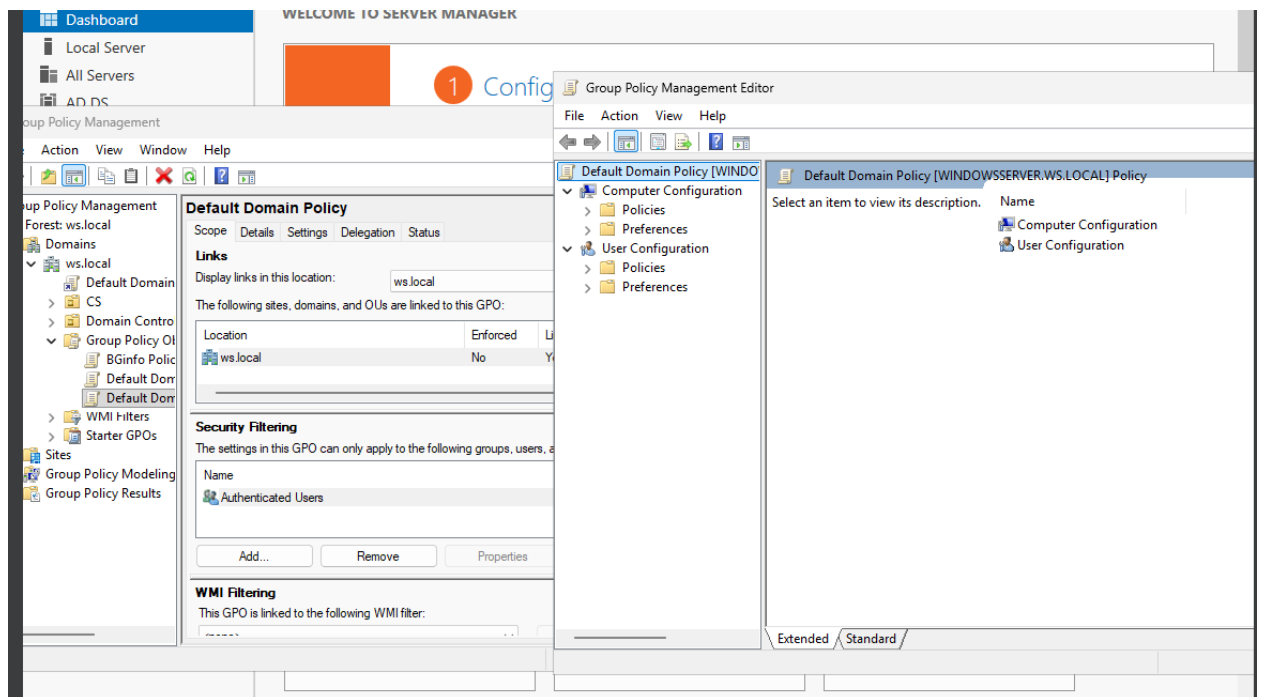
Next we add Bitlocker Drive encryption:



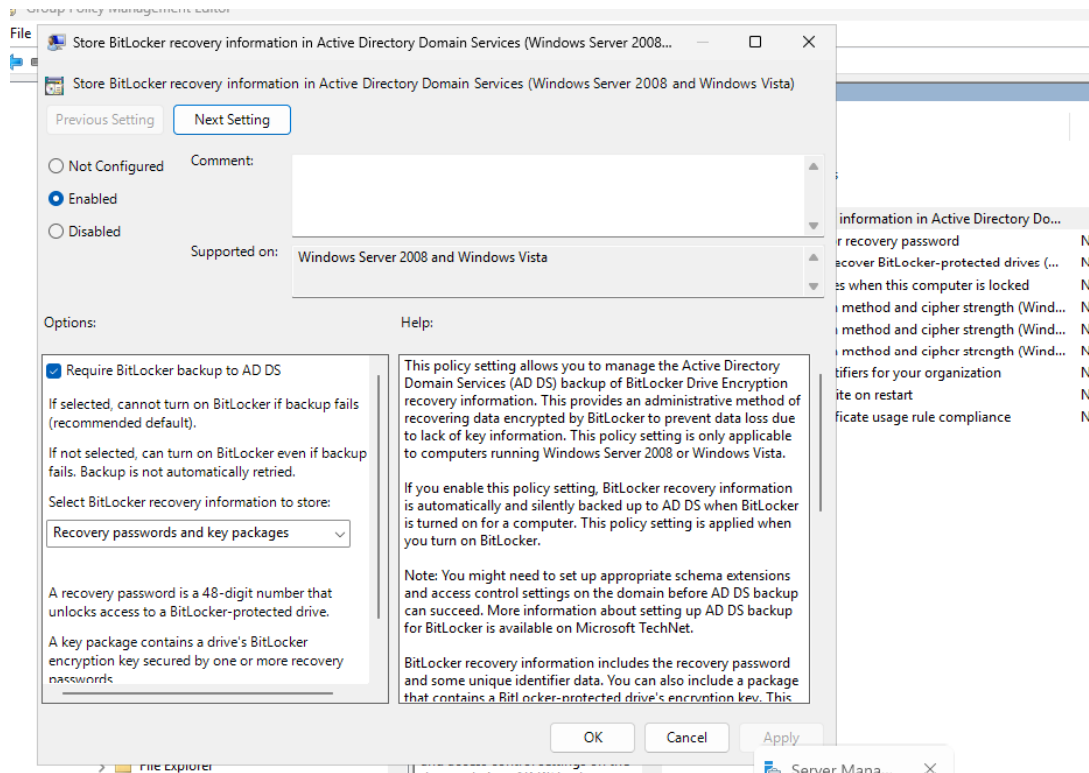
Make sure the restart button is pressed and install:



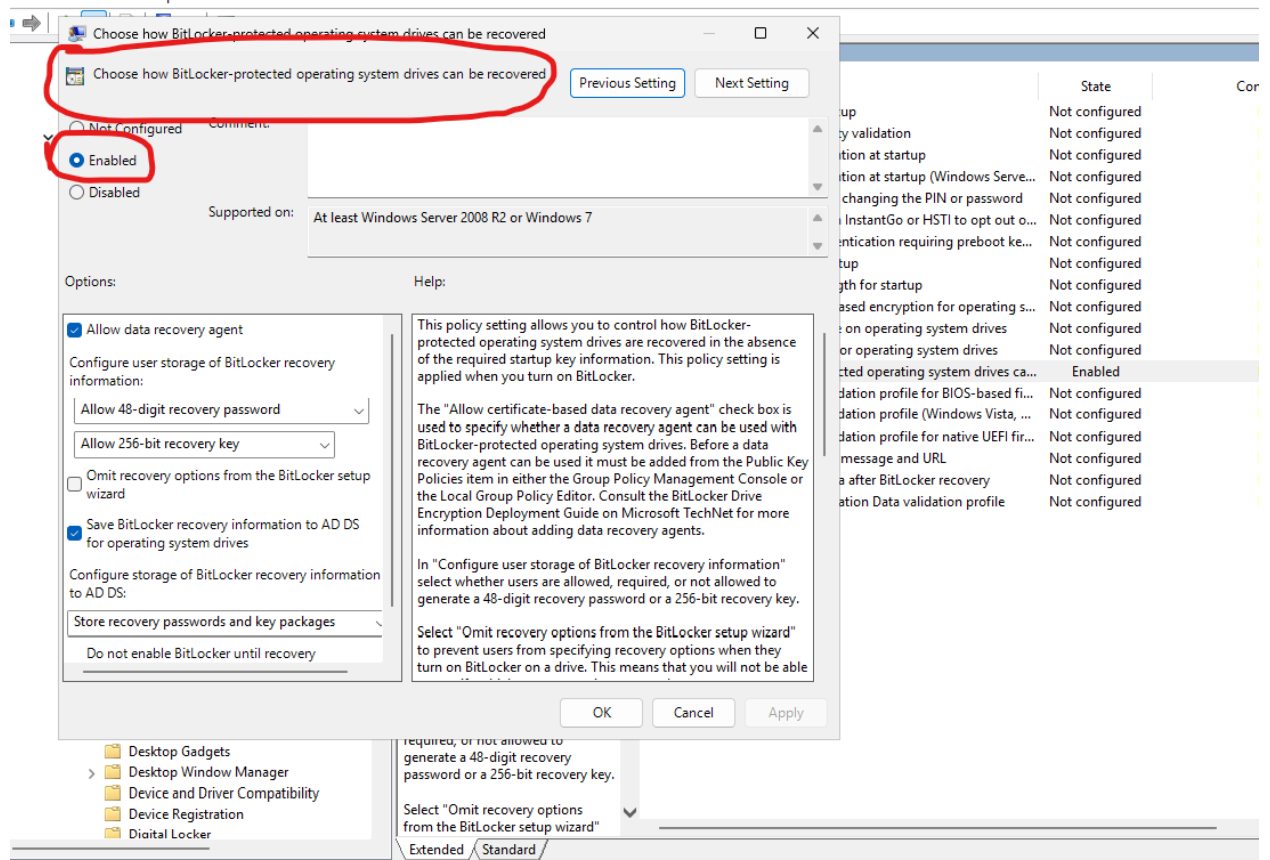
Then we go to Group Policy Management and Edit Default Domain Policy:



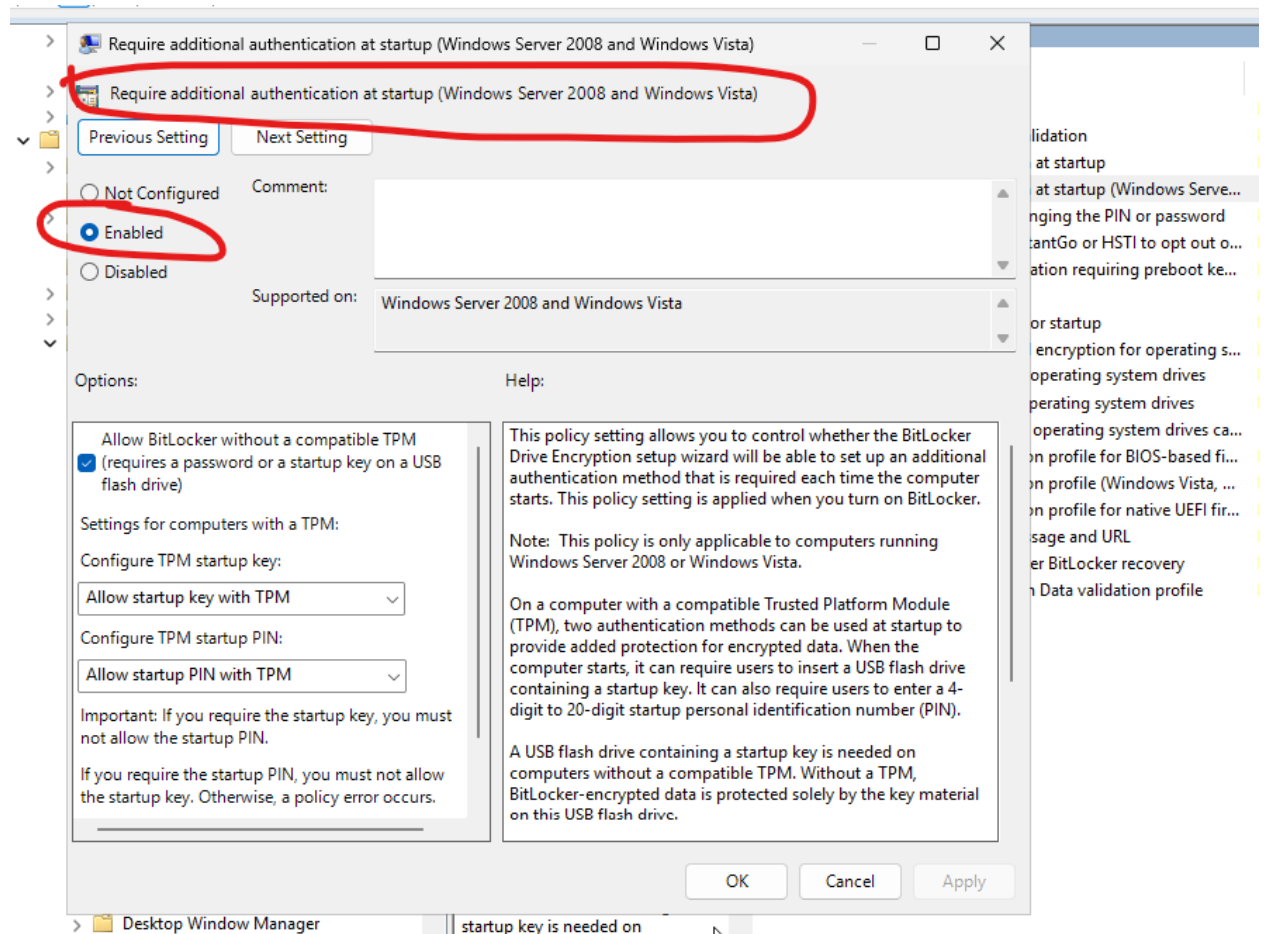
After We go to Administrative Templates and enable Store Bitlocker recovery:



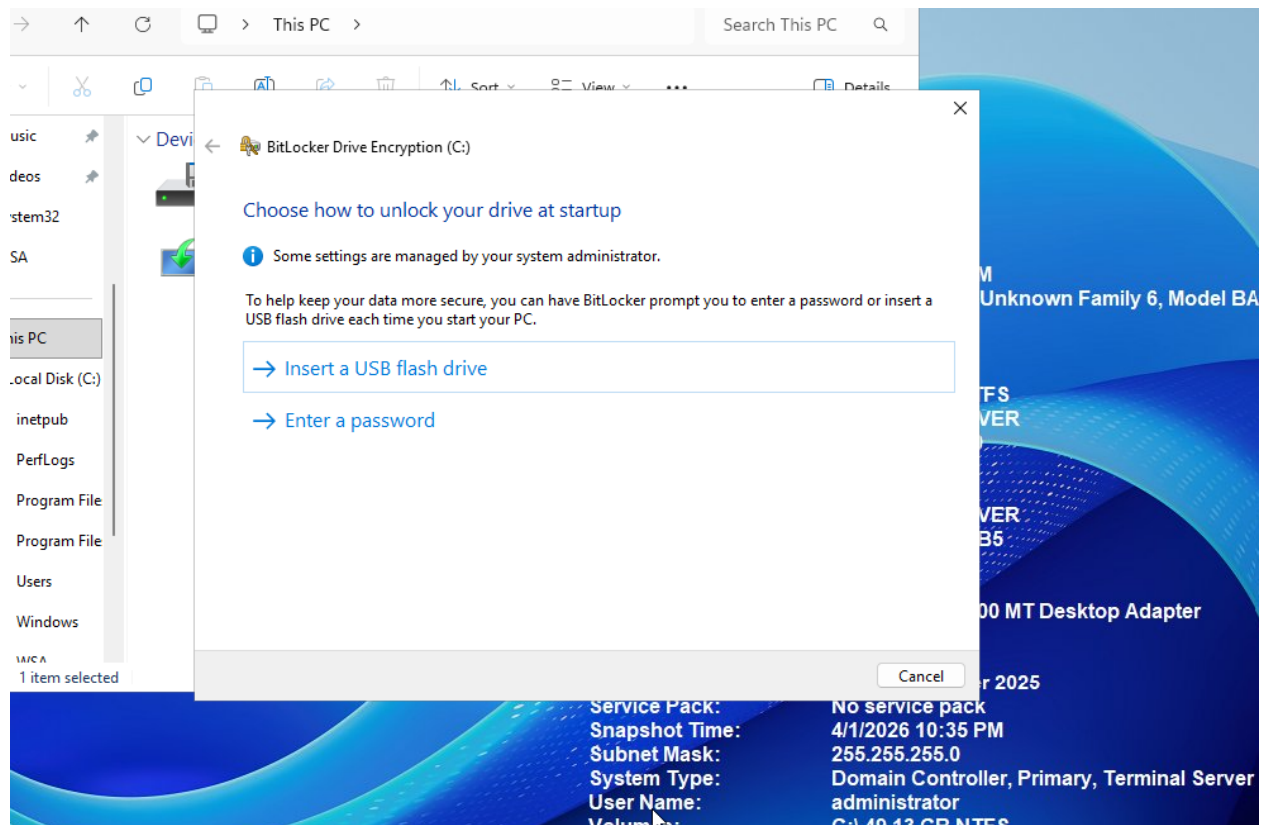
Then went to Operating system Drive section to enable again:



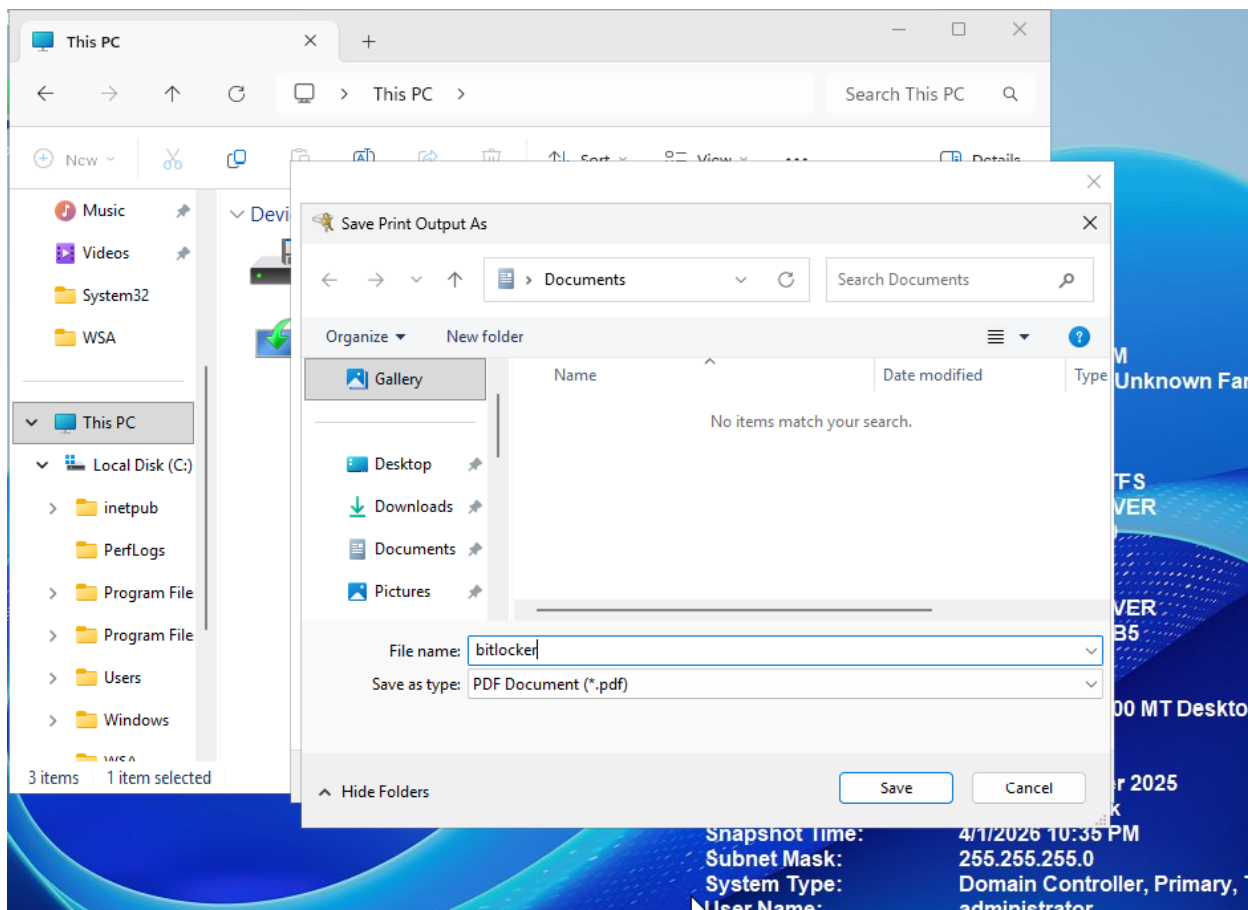
Also enabled:



Now we go to local disk and turn on Bitlocker and choose to enter passwd:



Next I create a password and print the recovery key:



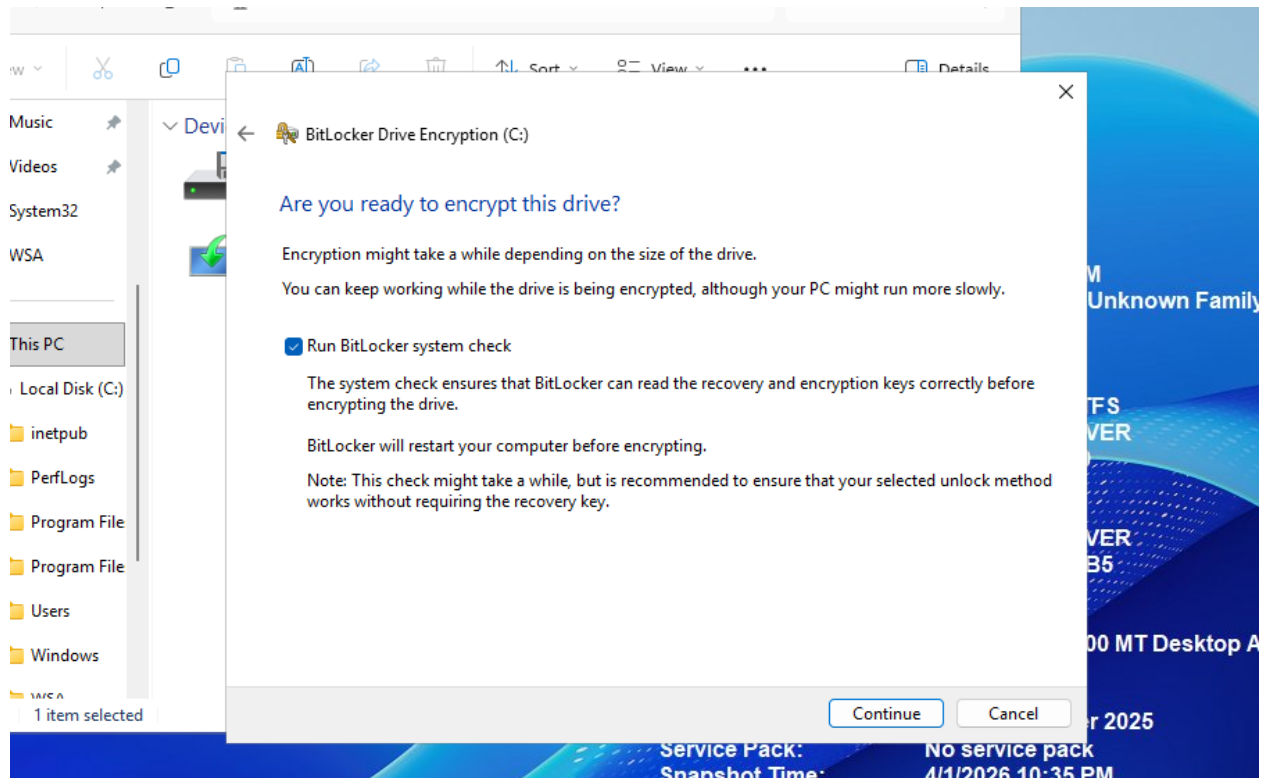
Choose how much of your drive to encrypt

If you're setting up BitLocker on a new drive or a new PC, you only need to encrypt the part of the drive that's currently being used. BitLocker encrypts new data automatically as you add it.

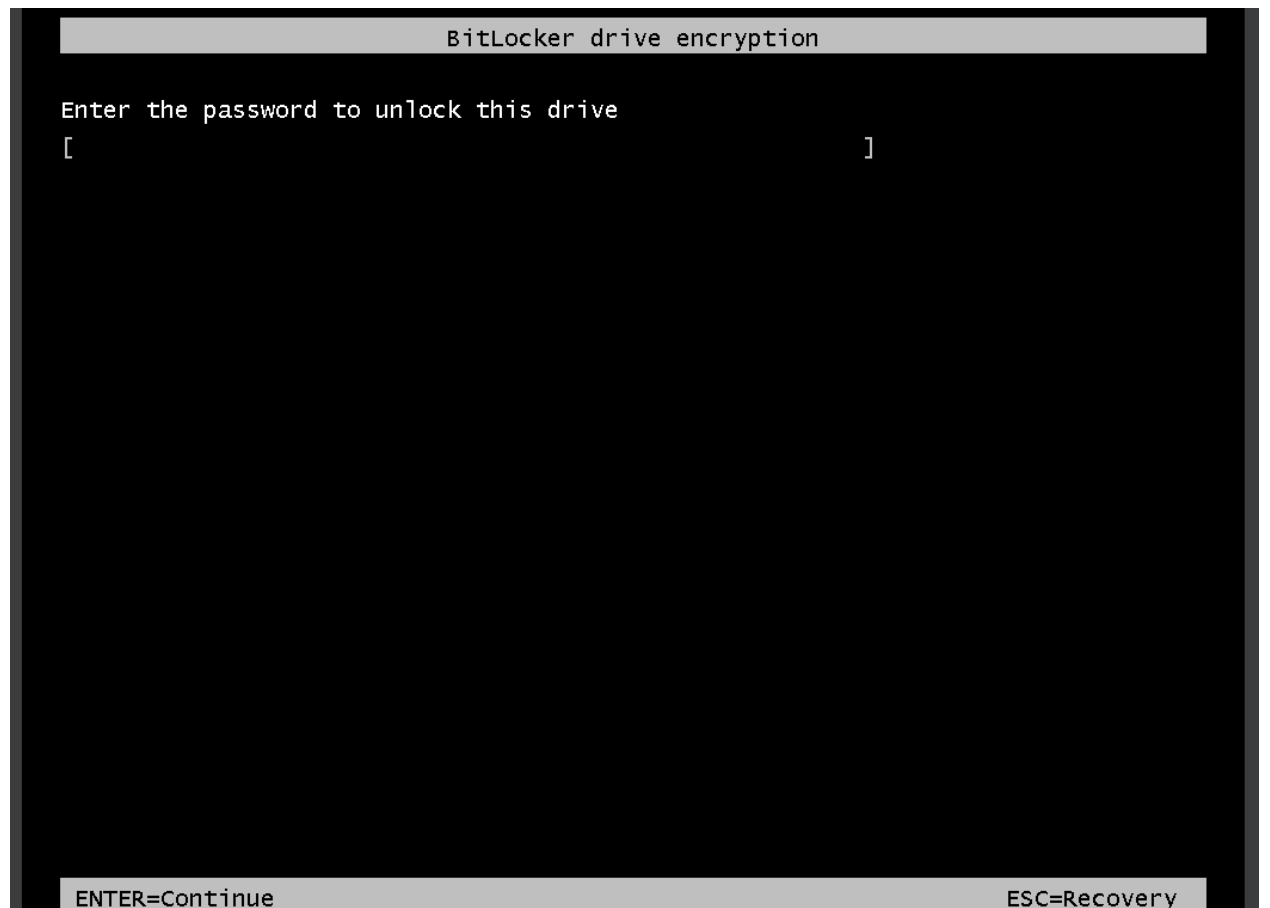
If you're enabling BitLocker on a PC or drive that's already in use, consider encrypting the entire drive. Encrypting the entire drive ensures that all data is protected—even data that you deleted but that might still contain retrievable info.

- ☒ Encrypt used disk space only (faster and best for new PCs and drives)

Getting ready and then we must restart the computer:



Then I enter password that I have set:



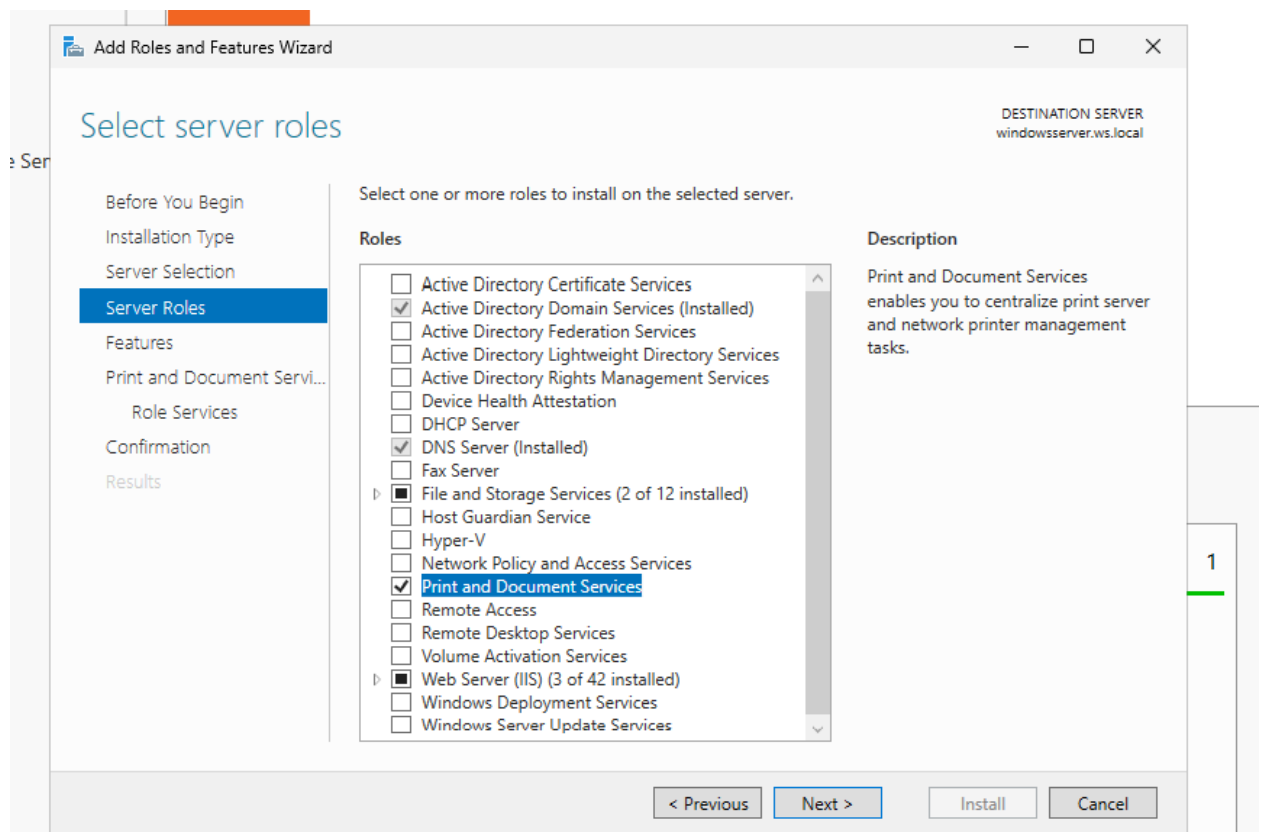
BitLocker is a Windows security feature that protects your data by encrypting your entire drive. Encryption means your files are turned into unreadable code so no one can access them without permission.

Its main purpose is to keep your data safe if your computer is lost, stolen, or accessed by someone else. Even if someone removes your hard drive and connects it to another computer, they won't be able to read your files without the correct password, PIN, or recovery key.

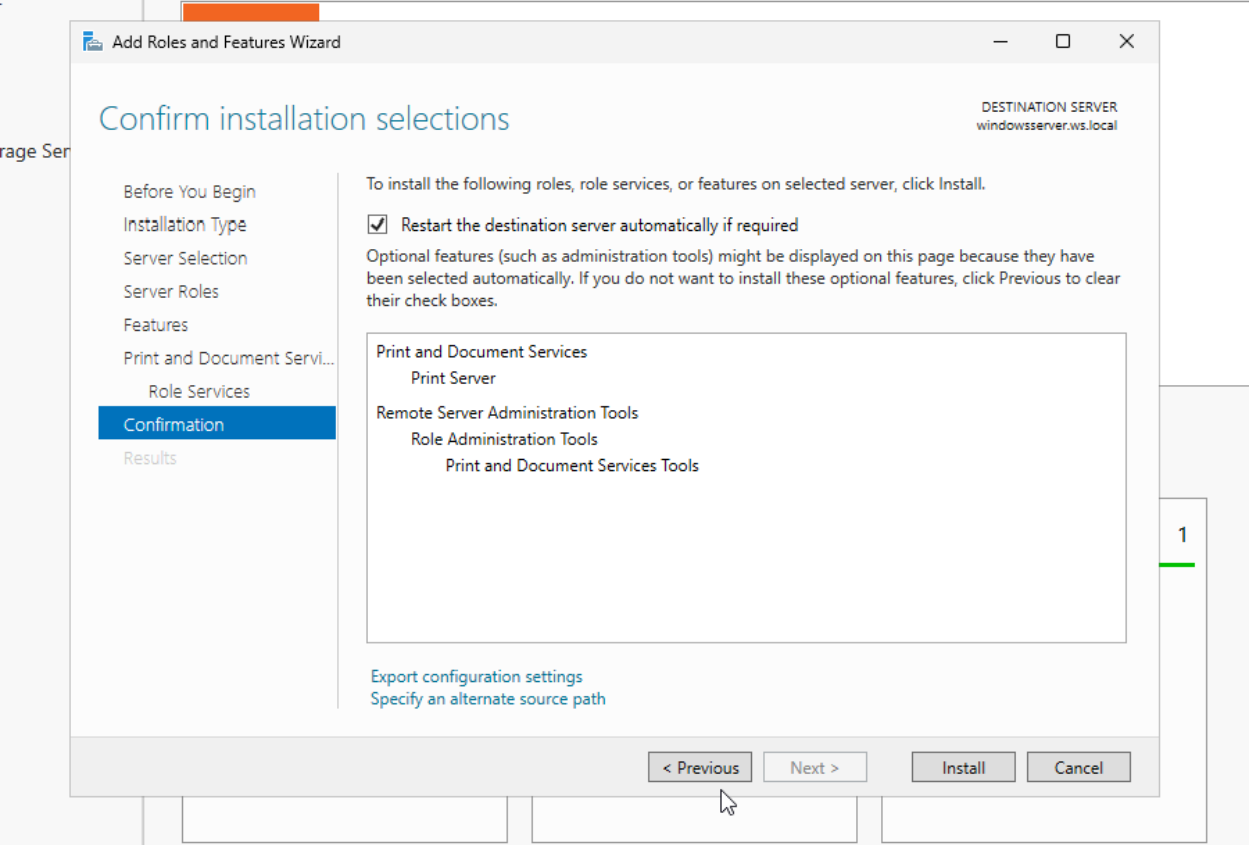
In simple terms, BitLocker locks your data so only authorized users can open and use it.

Part 3. Setup a Windows Print Server.

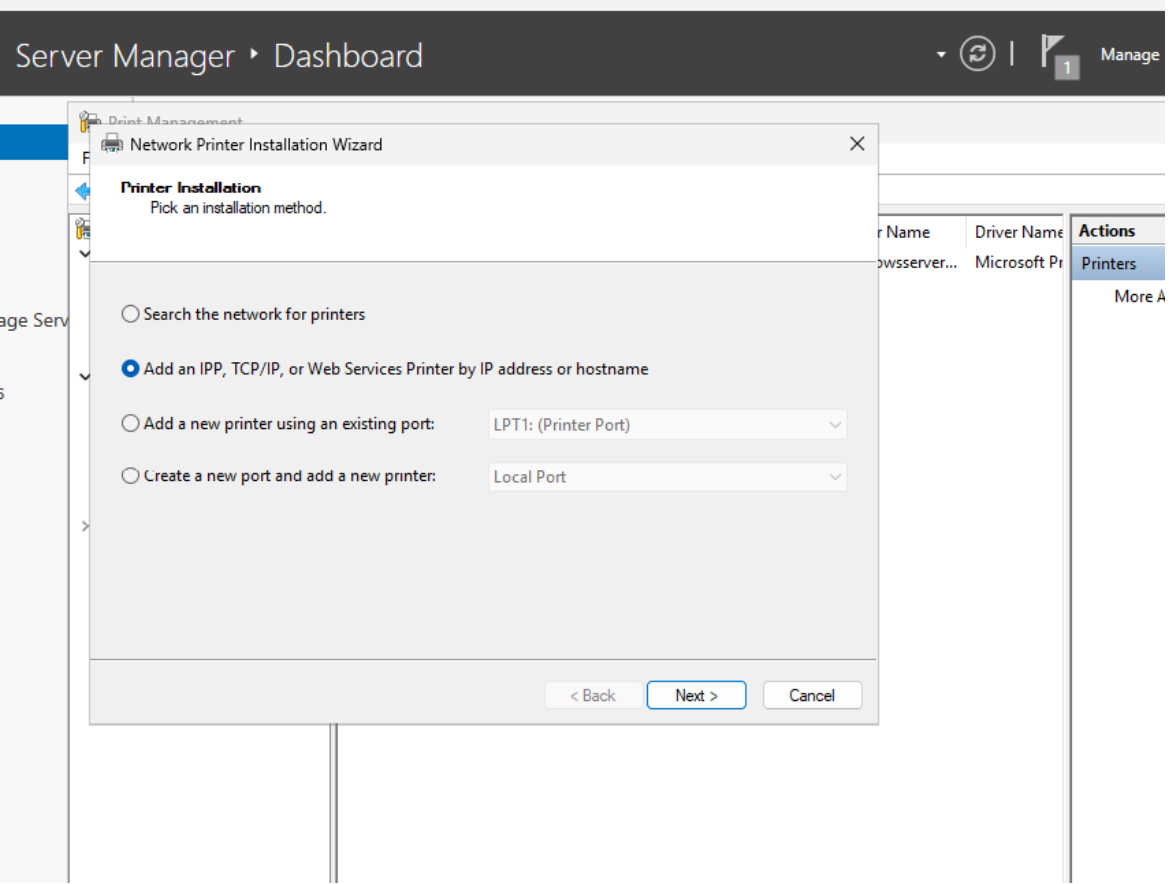
First, We go to Add Roles and features and tick a Print and Document Services:



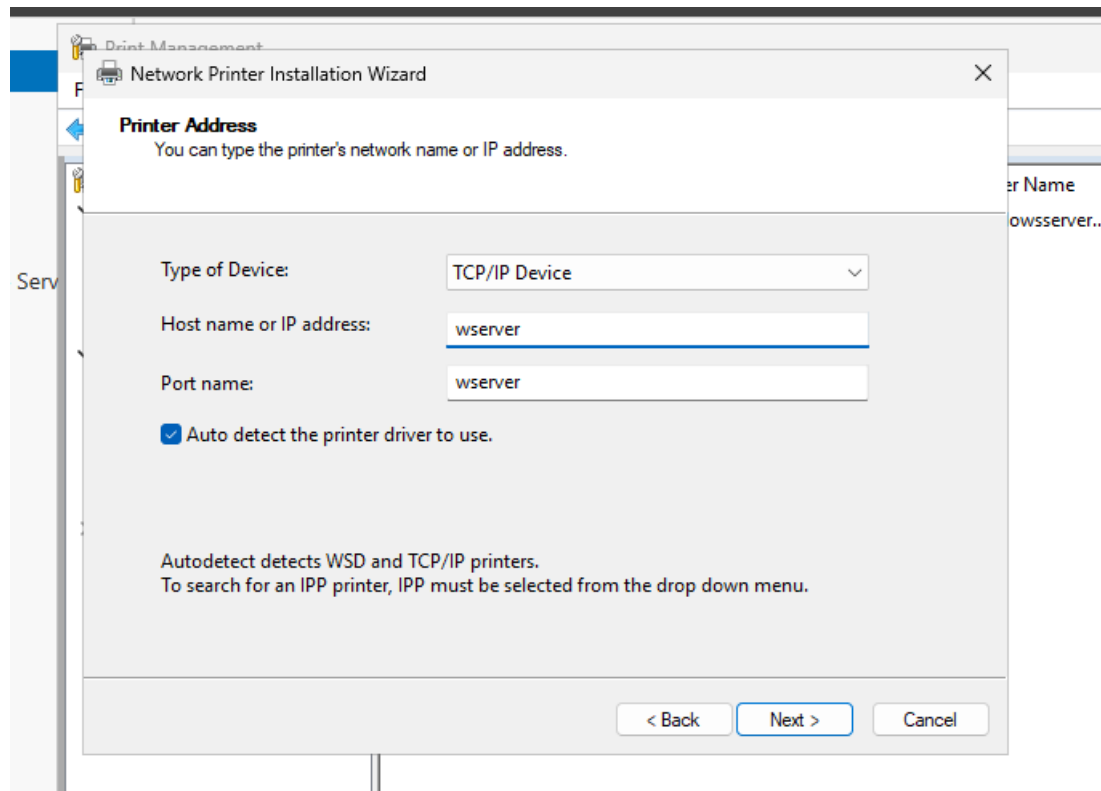
After we go to installation part:



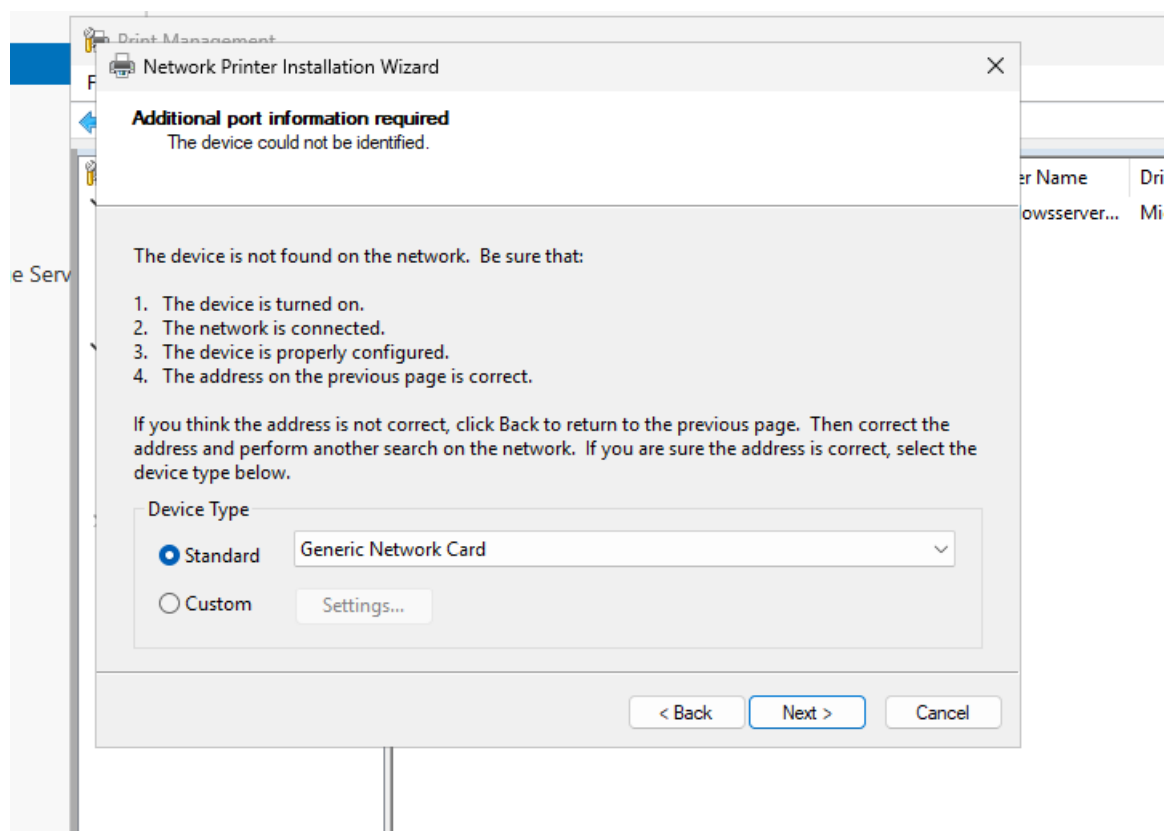
Then I go to Tools and choose Print Management and then add Printer:



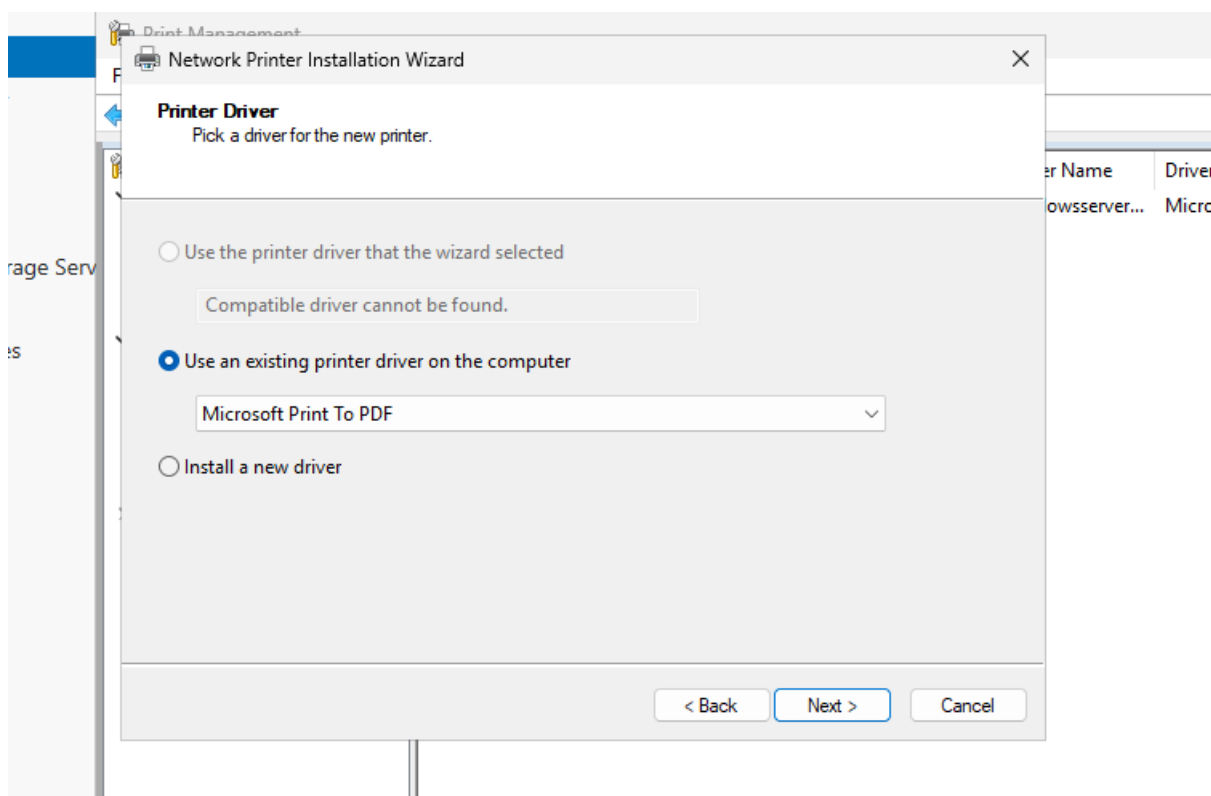
After I change type of device and write down info that is given in video:



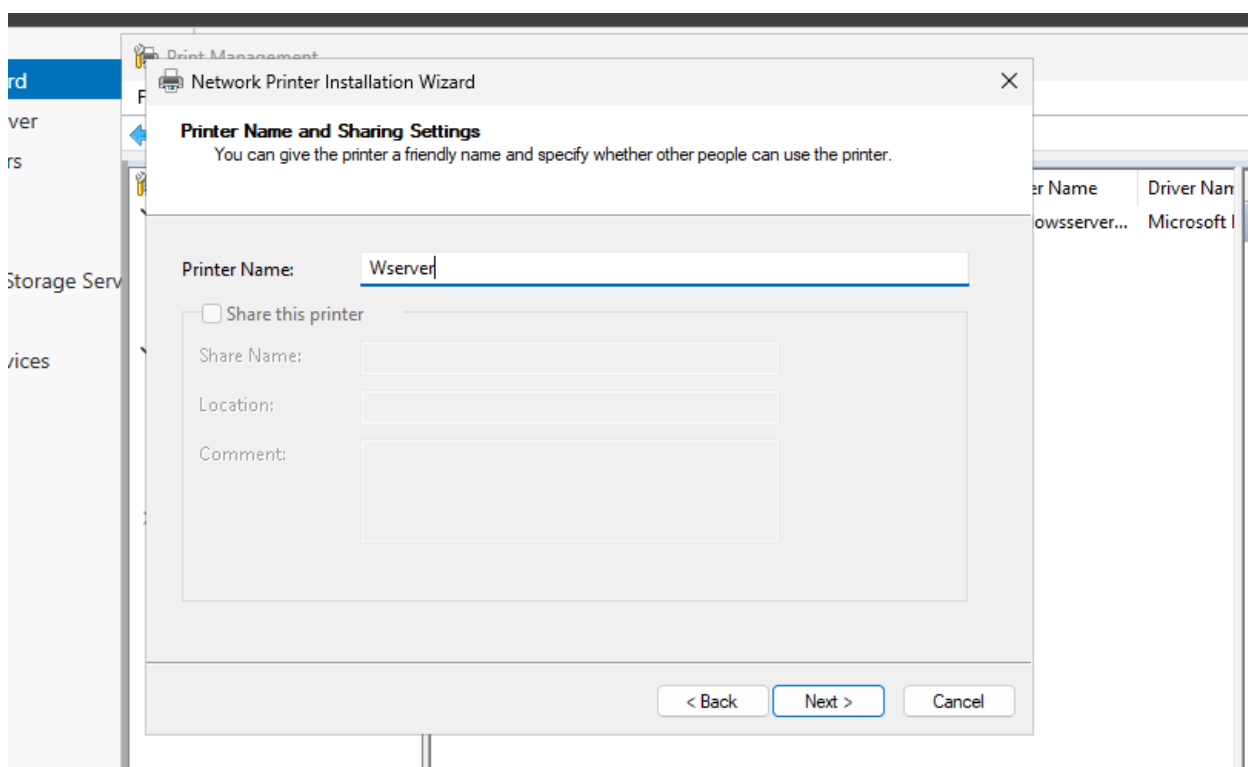
And then we go further next until we finish the installation:



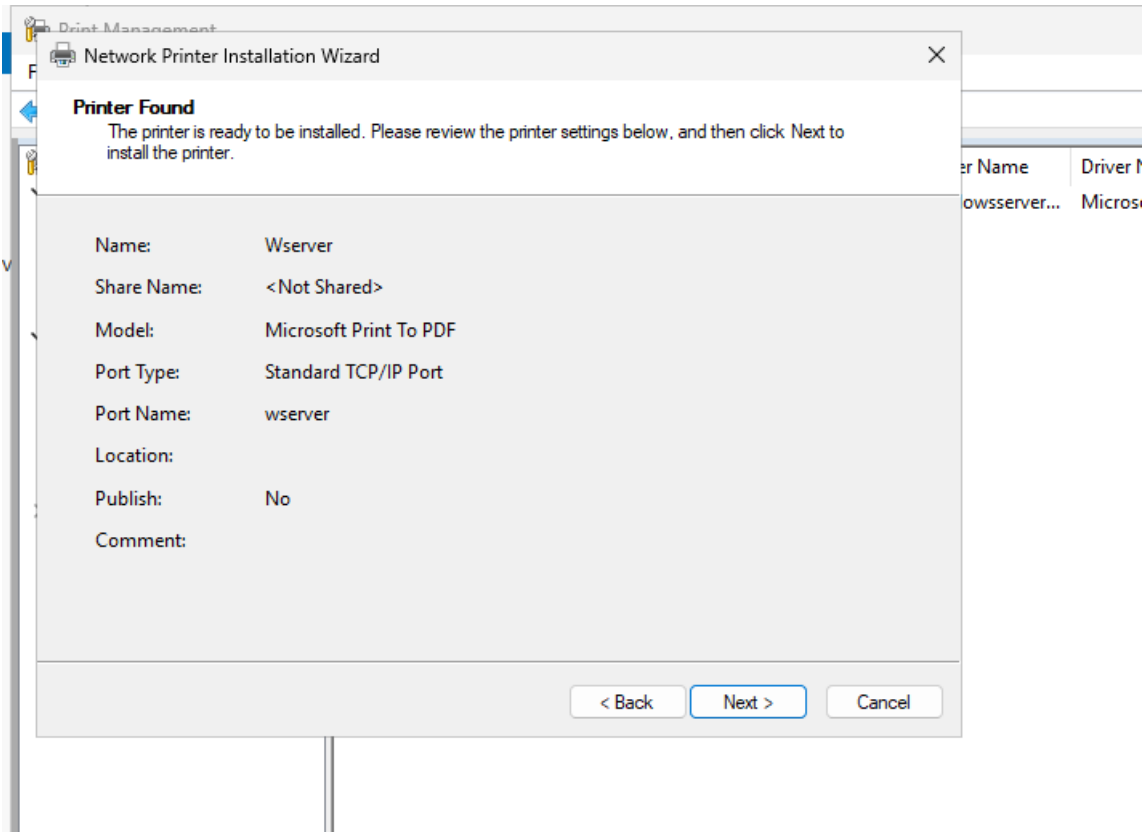
I decided to choose existing printer drivers:



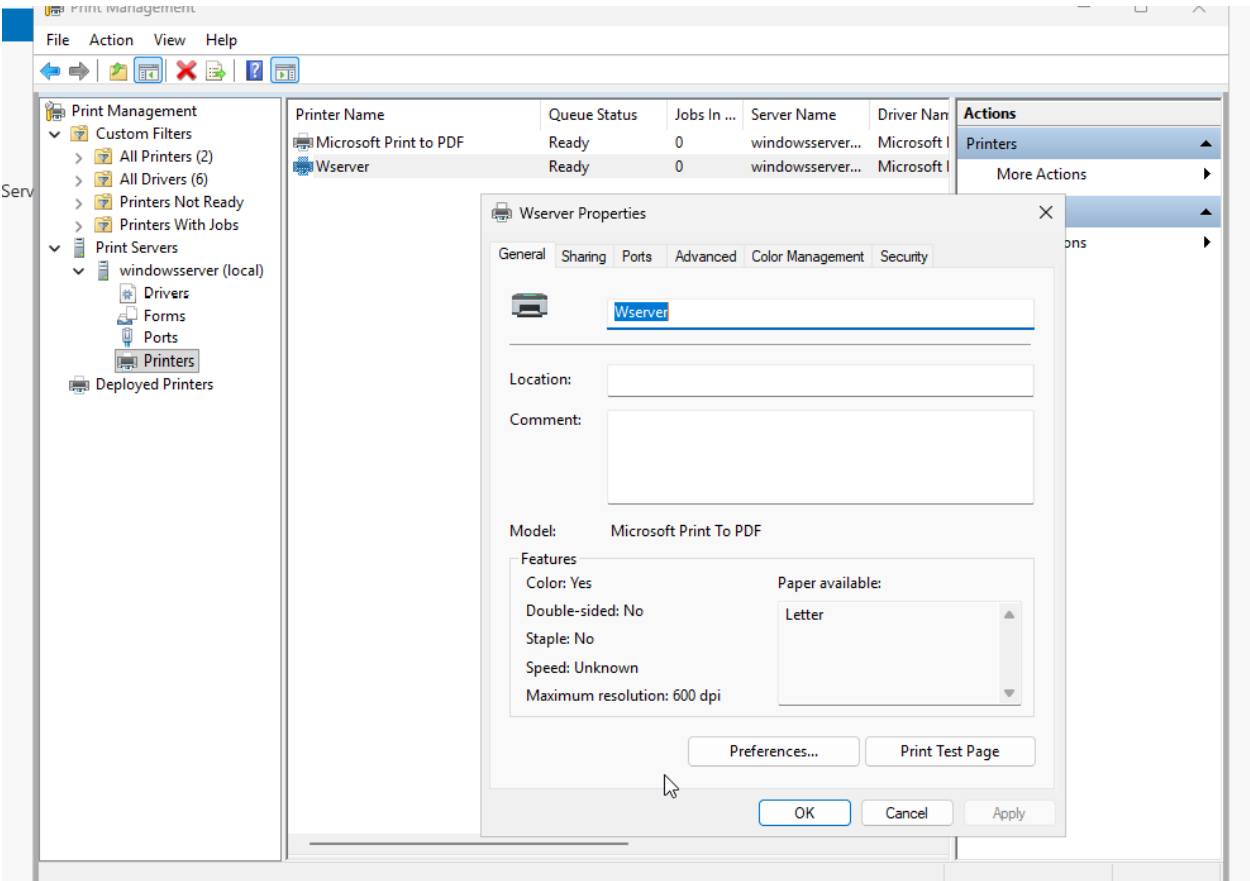
Then change a name to “Wserver” as in tutorial:



Final Settings:

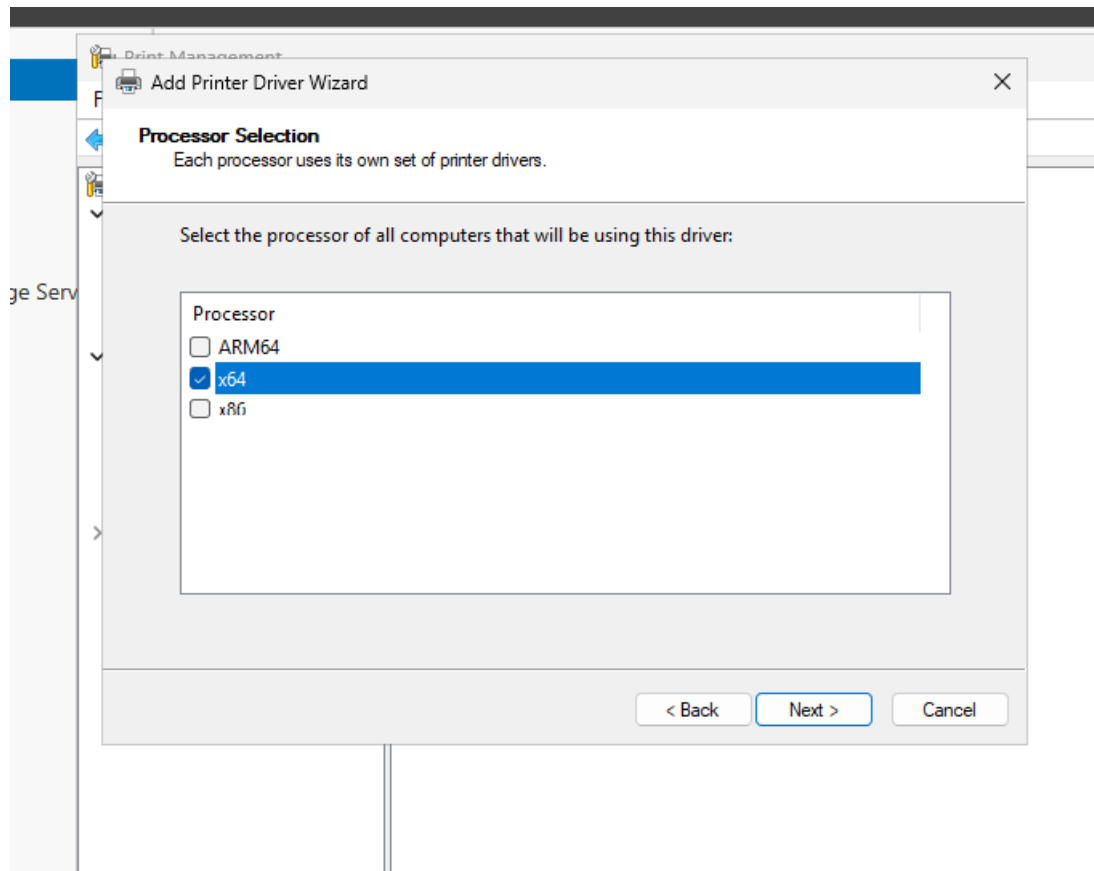


Finally, We have installed a printer Management and we can explore things with it:

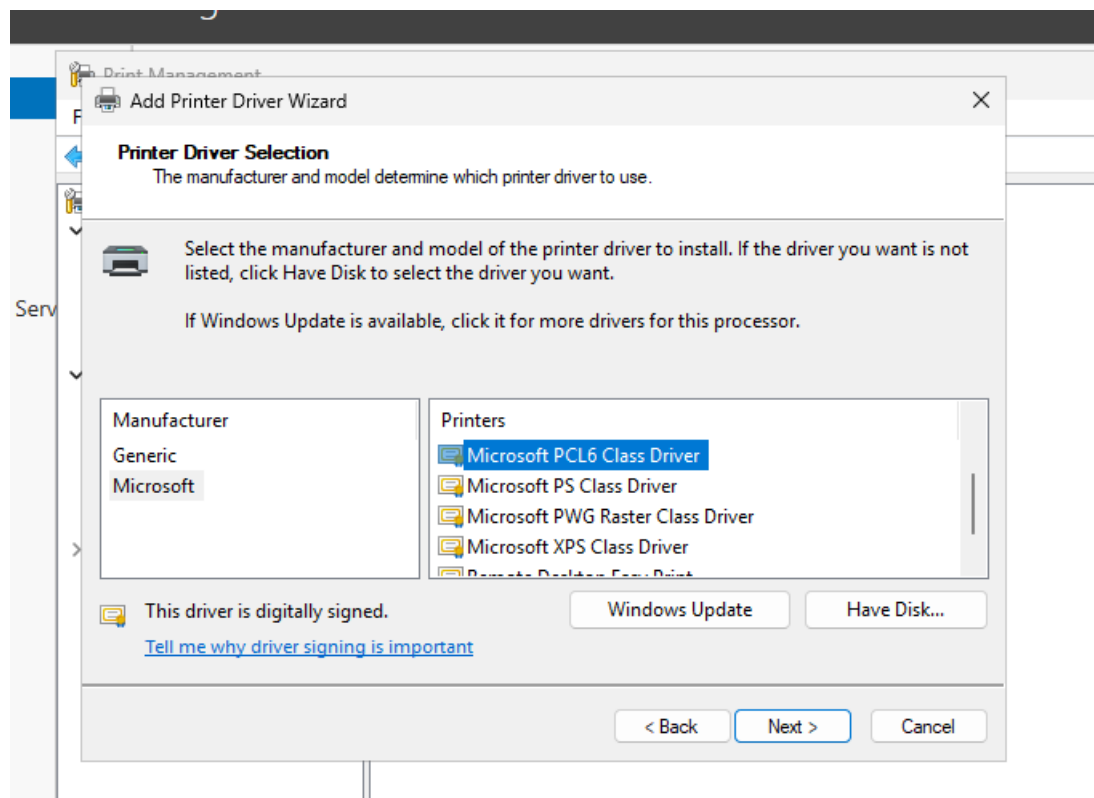


Part 4. Deploy printers using Group Policy

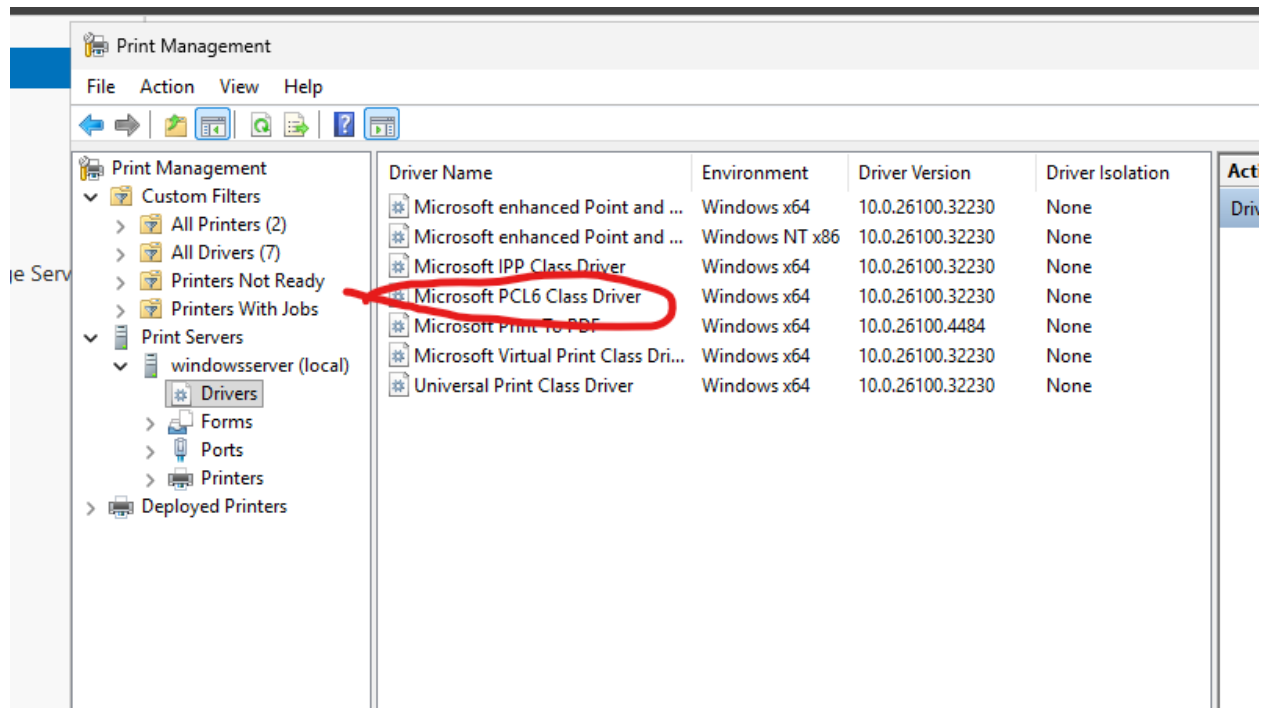
Firstly, the link that has been provided does not work, so I decided to use Universal driver for Printers:



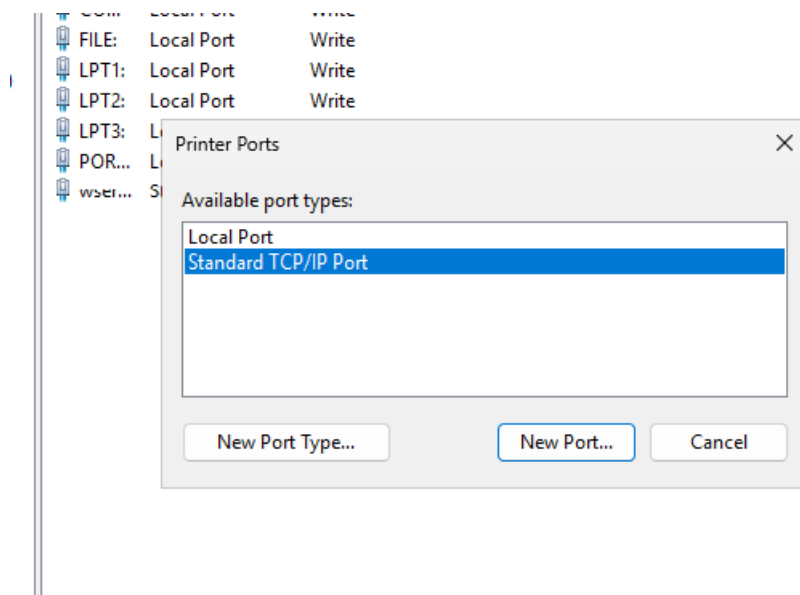
I have chosen a PCL 6 driver for simulation cause I don't have real printer:



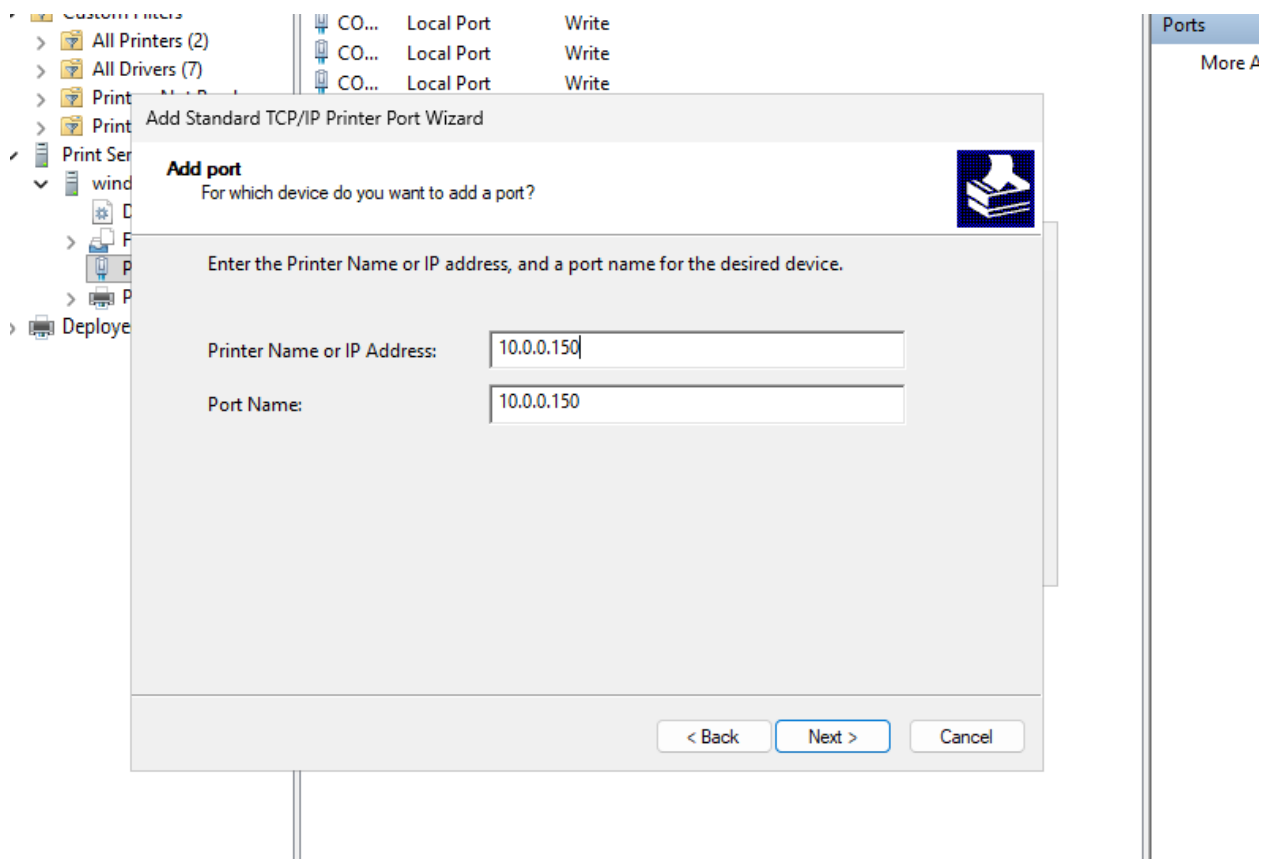
Here's my downloaded Driver:



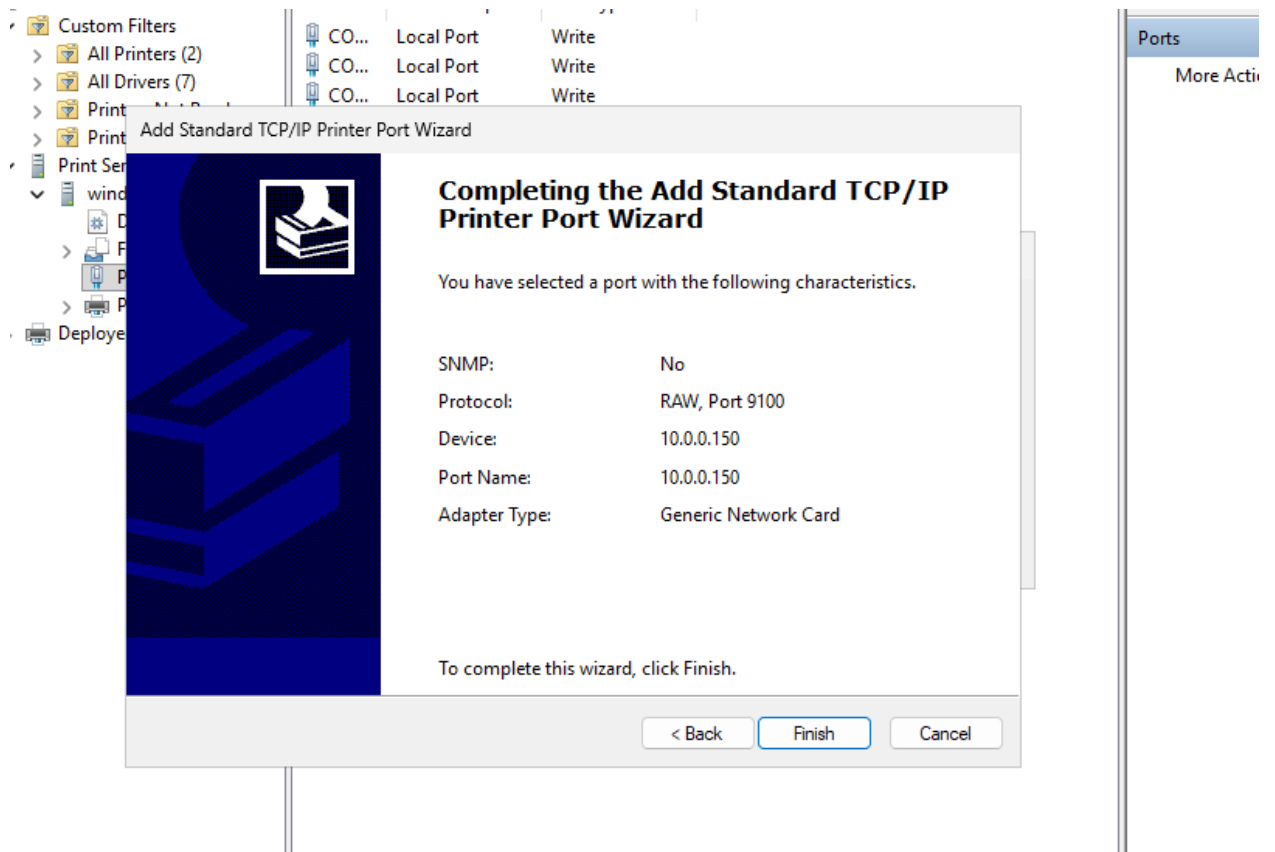
Now let's set up a port. Go to Ports -> right click -> Add Port... Select Standard TCP/IP Port -> New Port -> Next.



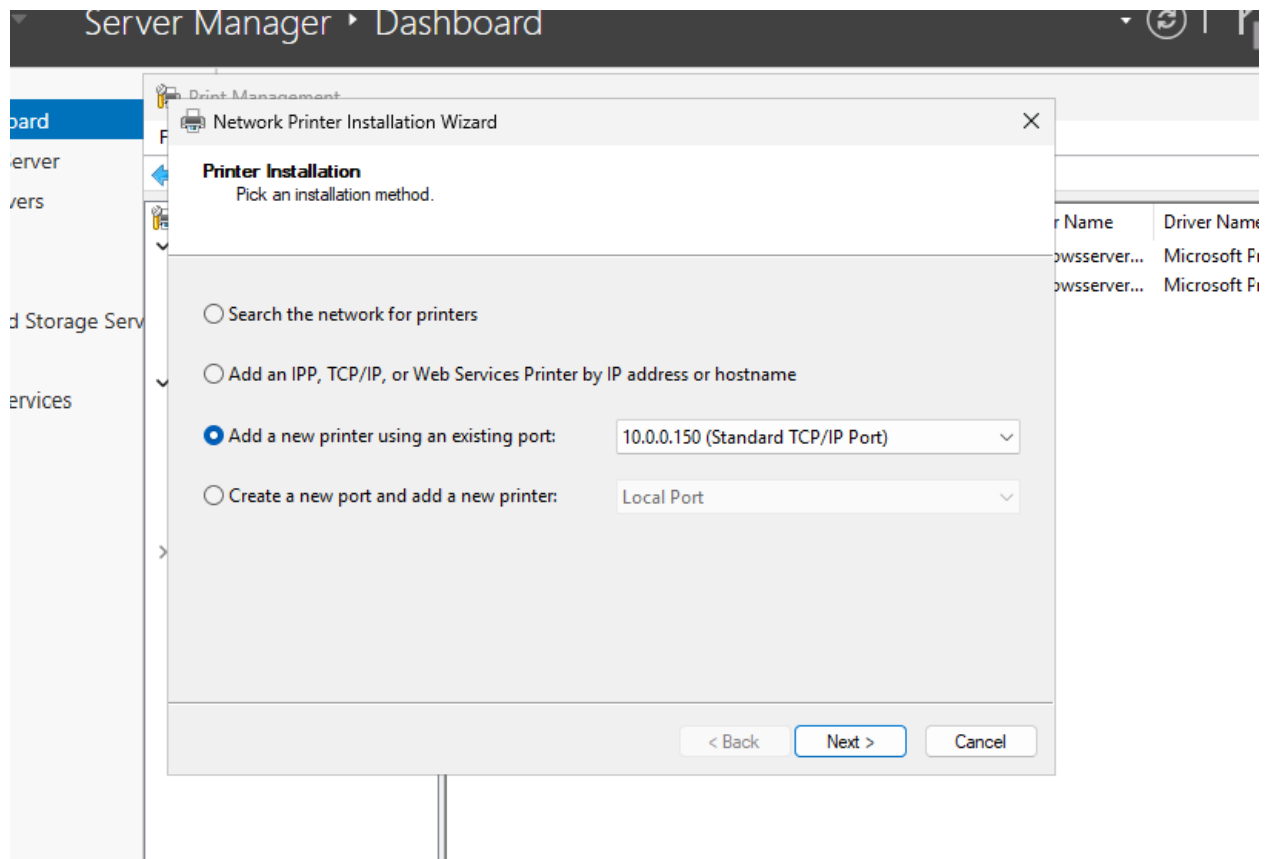
Enter the IP address of the printer.



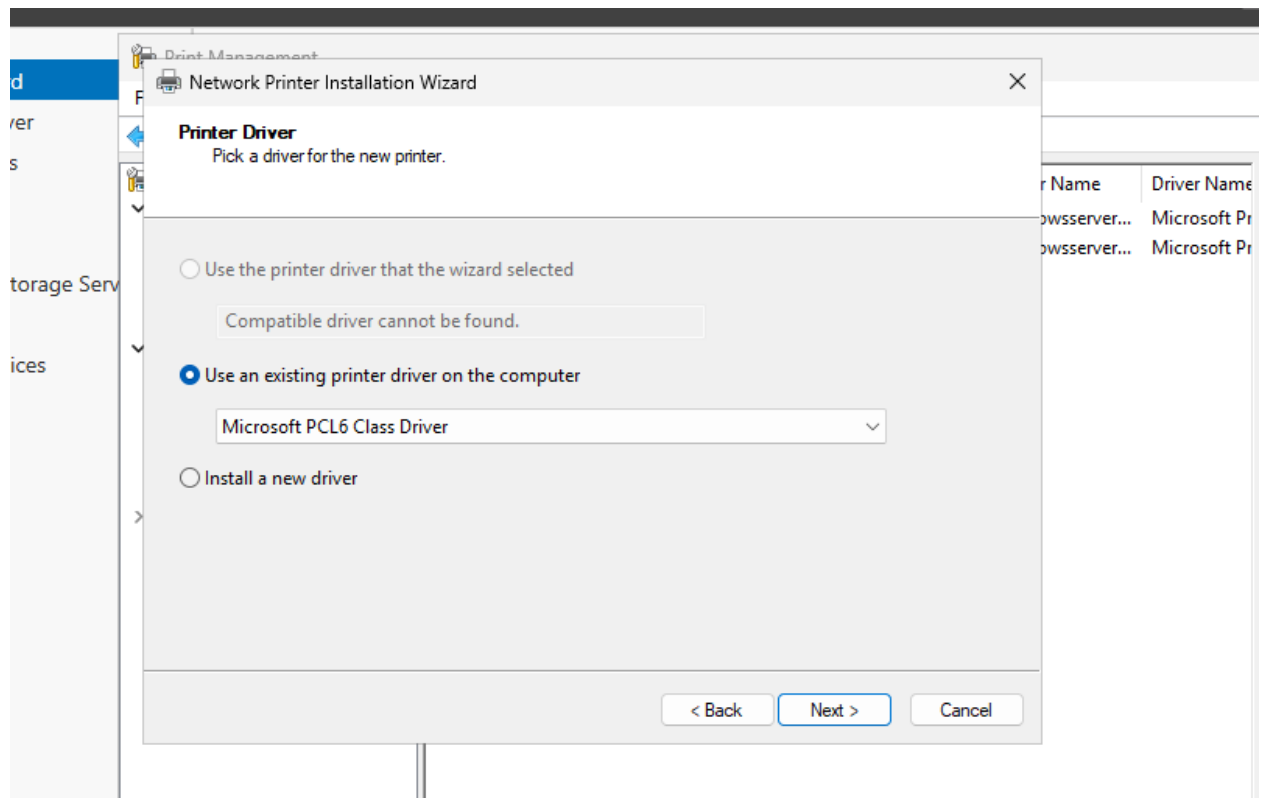
Press Next -> Finish-> Close.



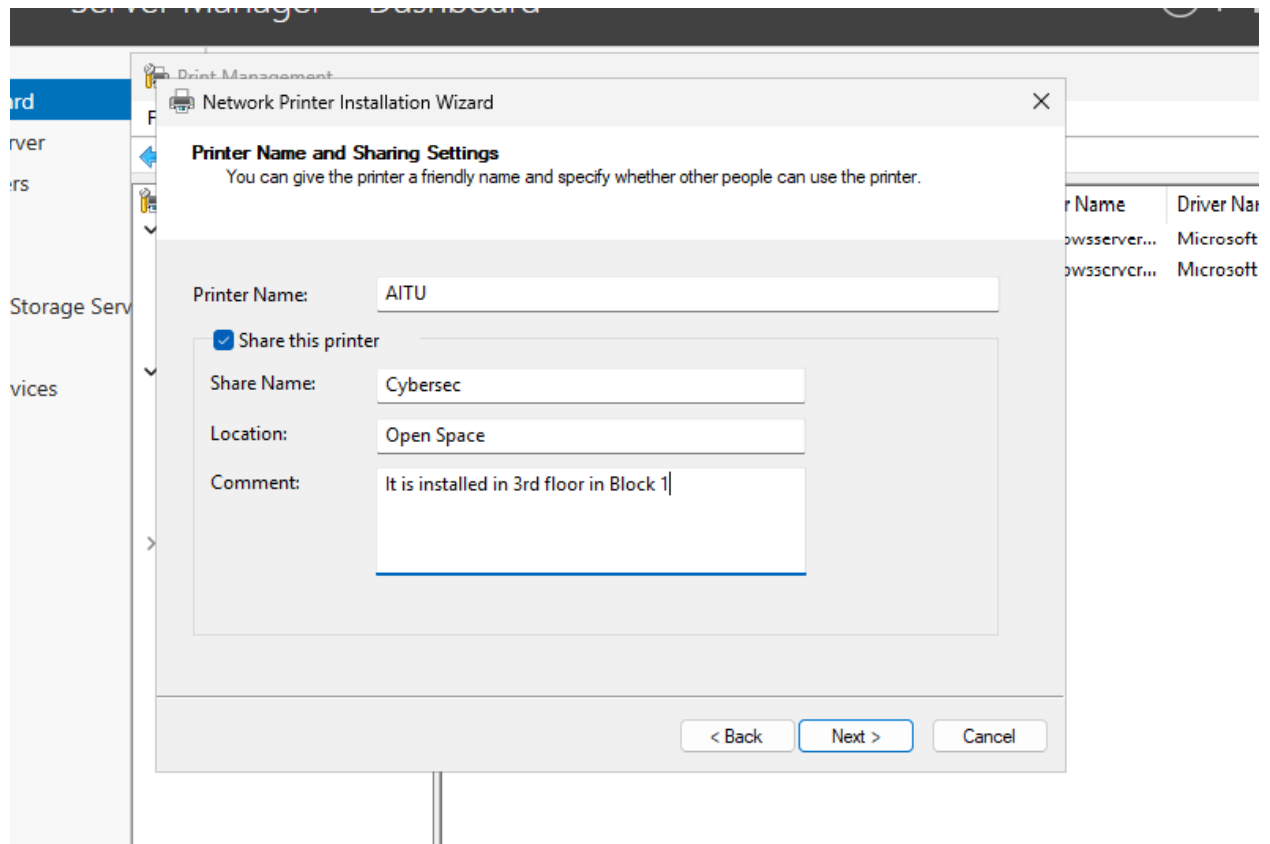
As we have the driver and port set up, we can now go to create the printer. Go to Printers -> Add Printer Select Add a new printer using an existing port, and choose the previously created port.



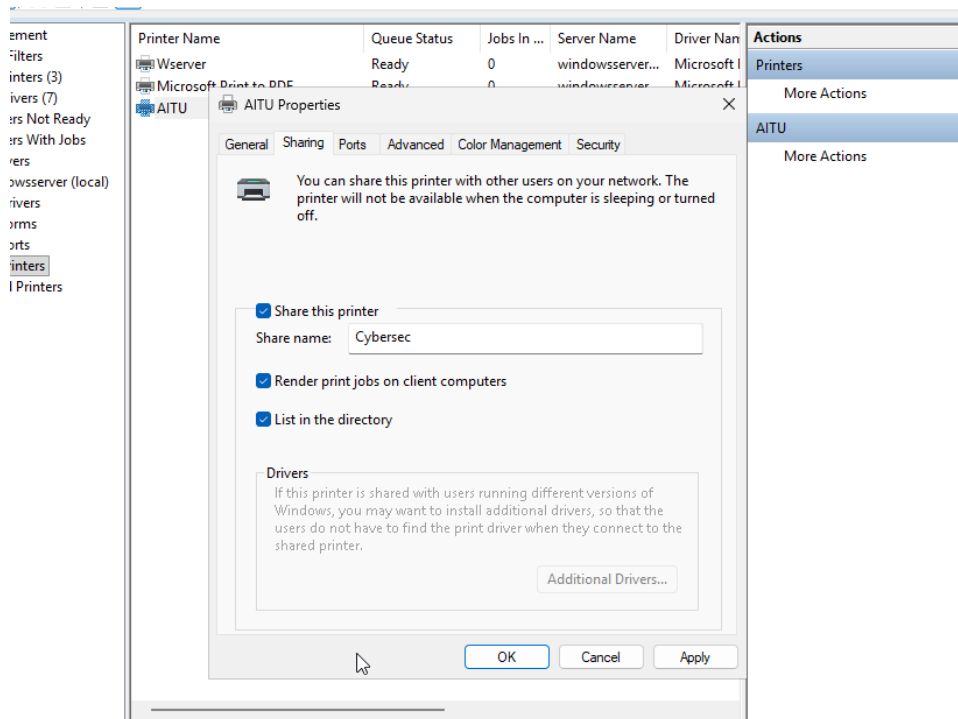
Use an existing printer driver which we downloaded and press Next



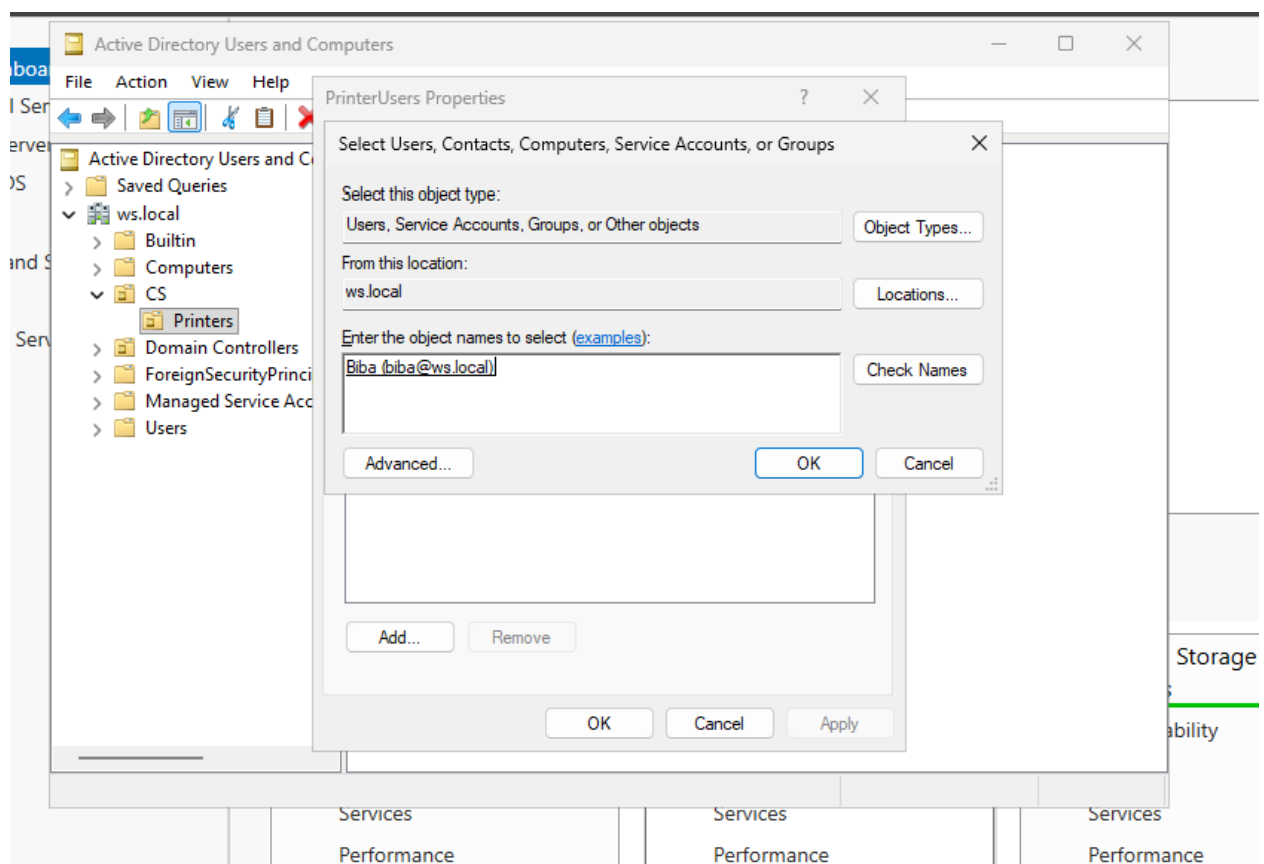
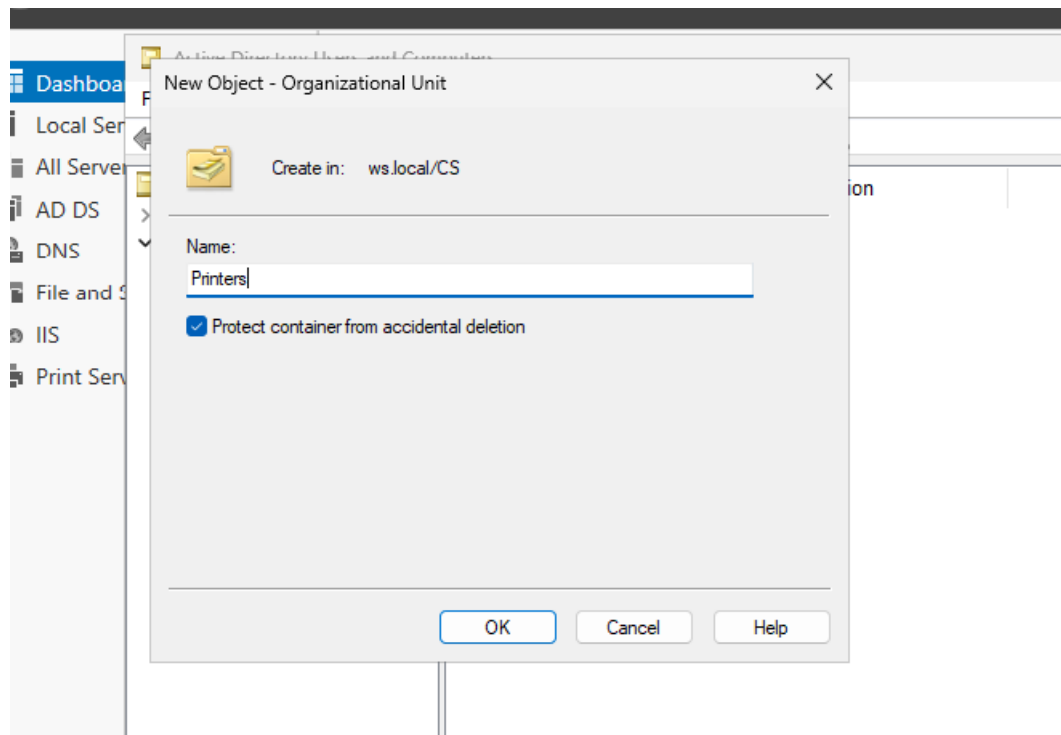
Change the name of your printer as you wish. Tick Share this printer, and write your own Share Name, Location (any place in the probable company, like Main Hall, Office 2 nd floor, or any other place), and the comment to clarify where exactly this printer is installed) -> Next-> Finish.



Now let's share the printer. Go to Printer and choose the newly installed printer. Right click on it -> Properties-> Sharing. Make sure that it was shared. Additionally, select List in the directory.

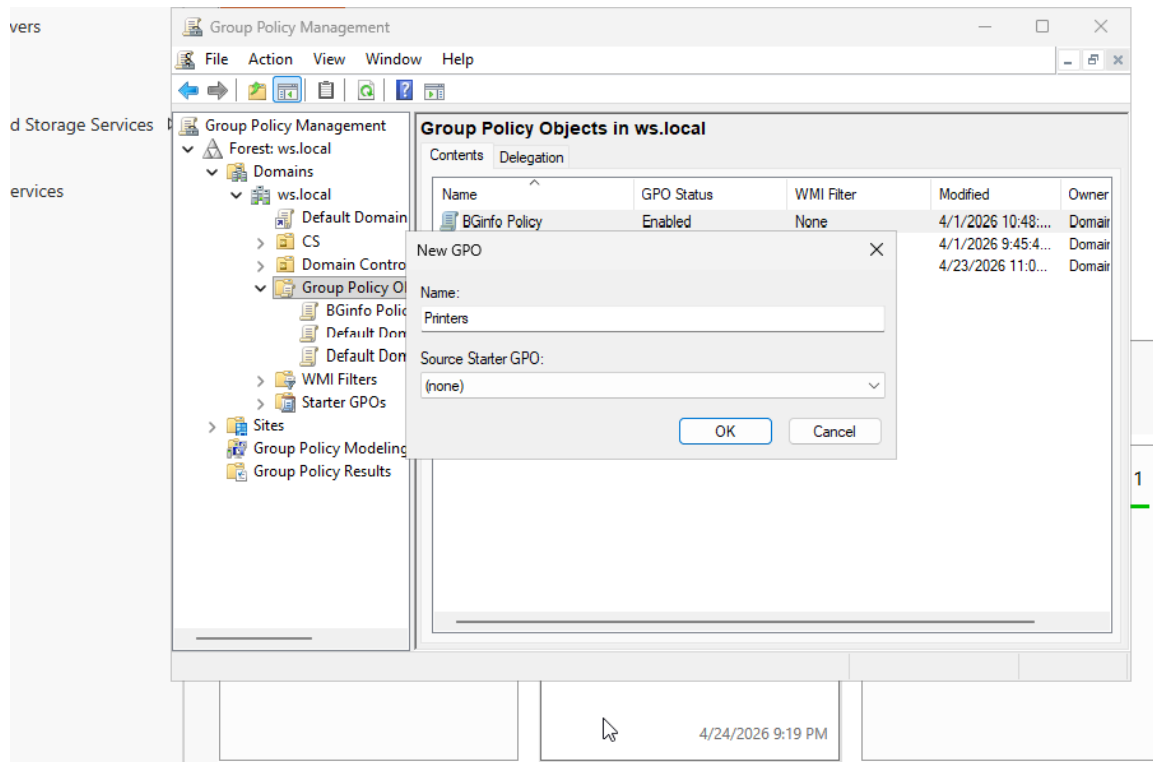


Now the printer is ready to be deployed. Go to Active Directory Users and Computers. Choose the CS organizational unit and create new organizational unit there with the name Printers. In the created OU create new Group and name it as you wish.

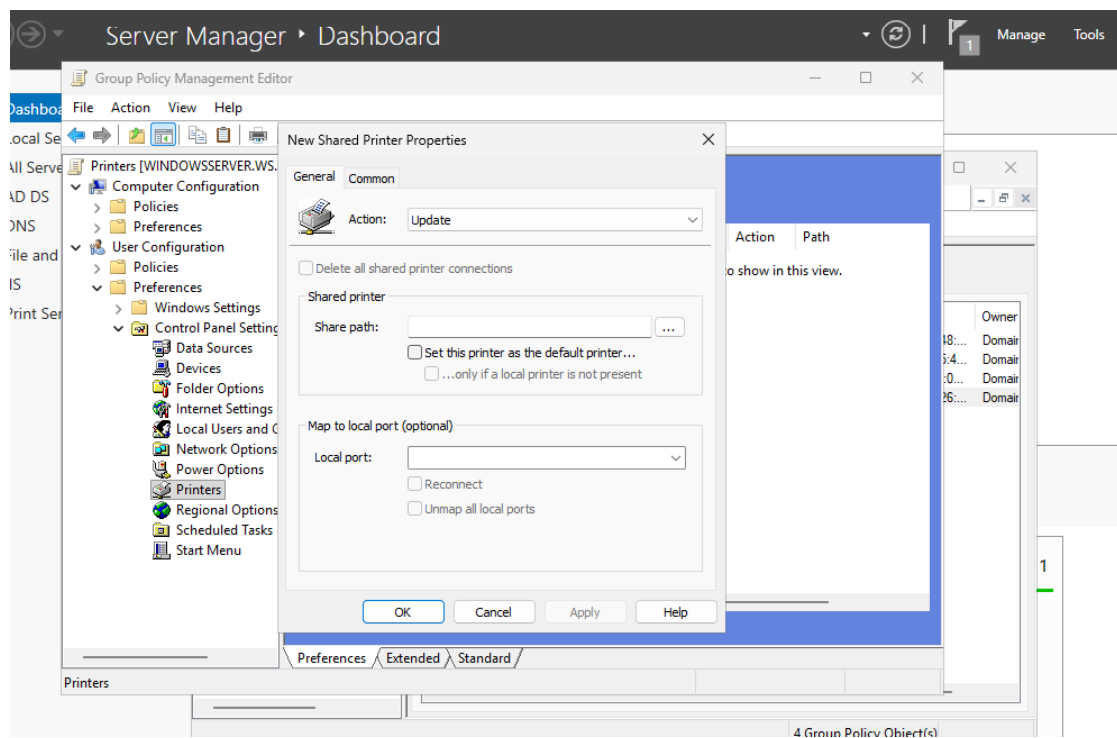


Now we will create a group policy to deploy the printer to anyone within the security group.

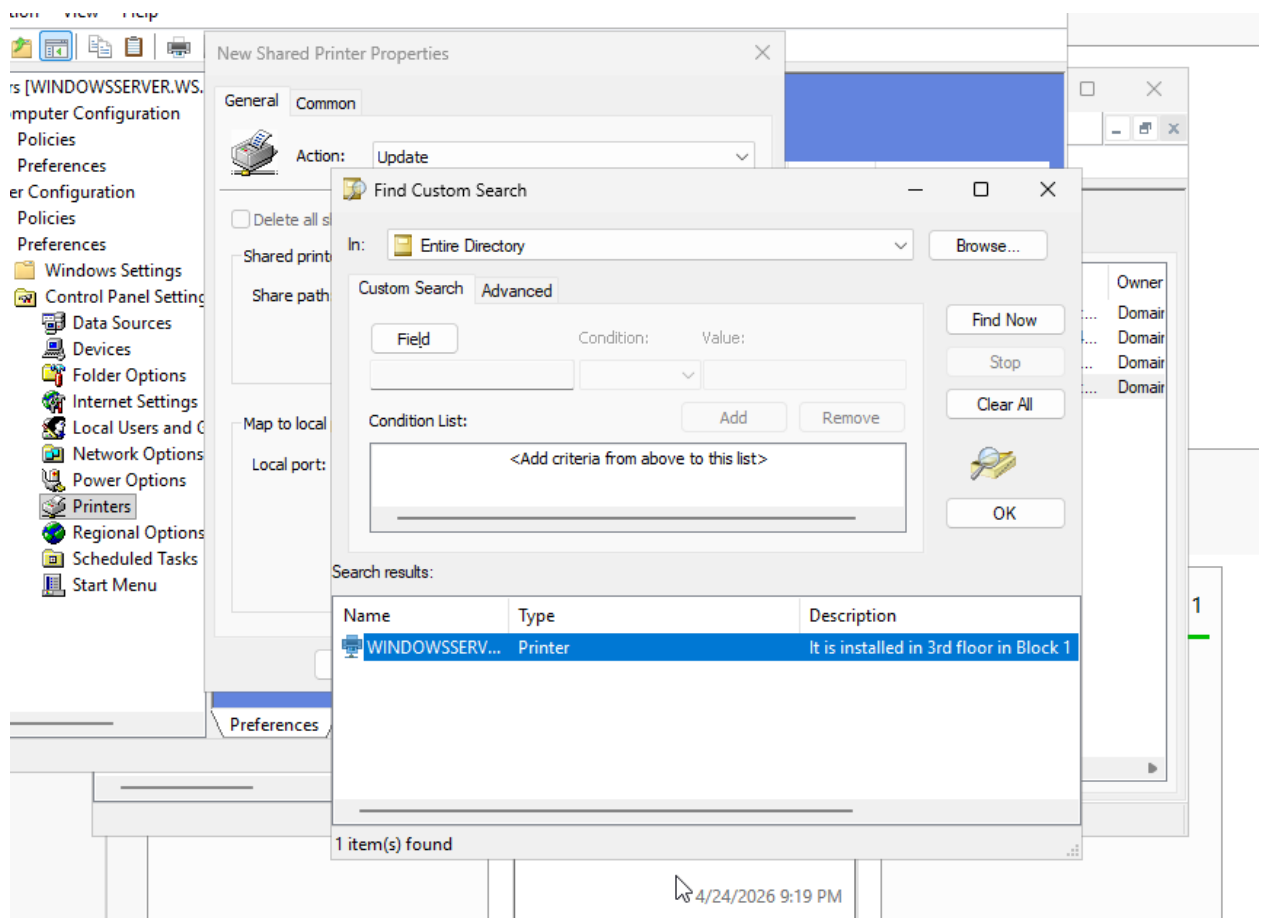
Go to the Group Policy Management-> Group Policy Objects-> right click -> New-> name it Printers.



In the list of available GPOs right click on Printers and then Edit. Go to User Configuration -> Preferences-> Control Panel Settings -> Printers. Right click -> New -> Shared Printer.

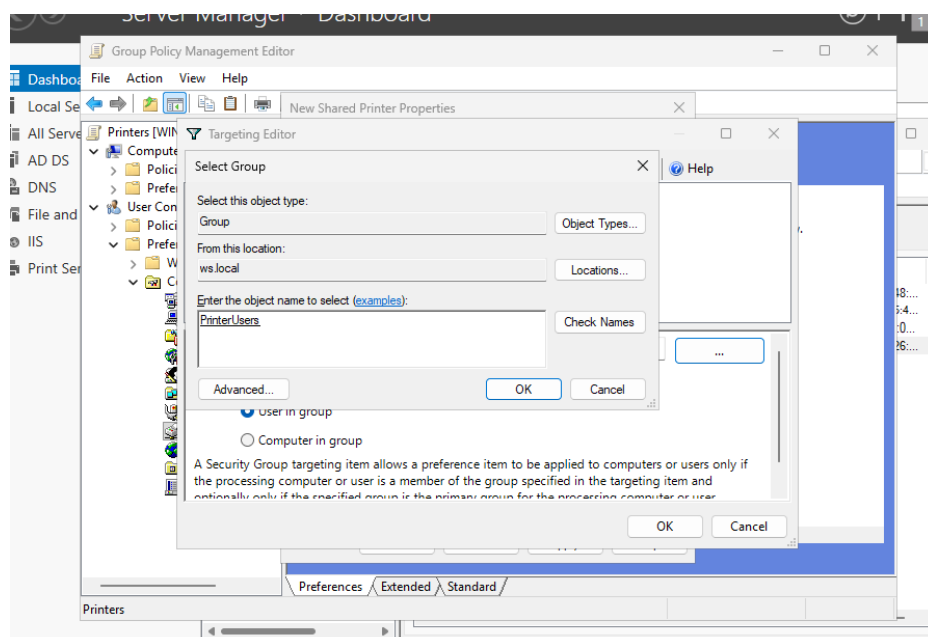


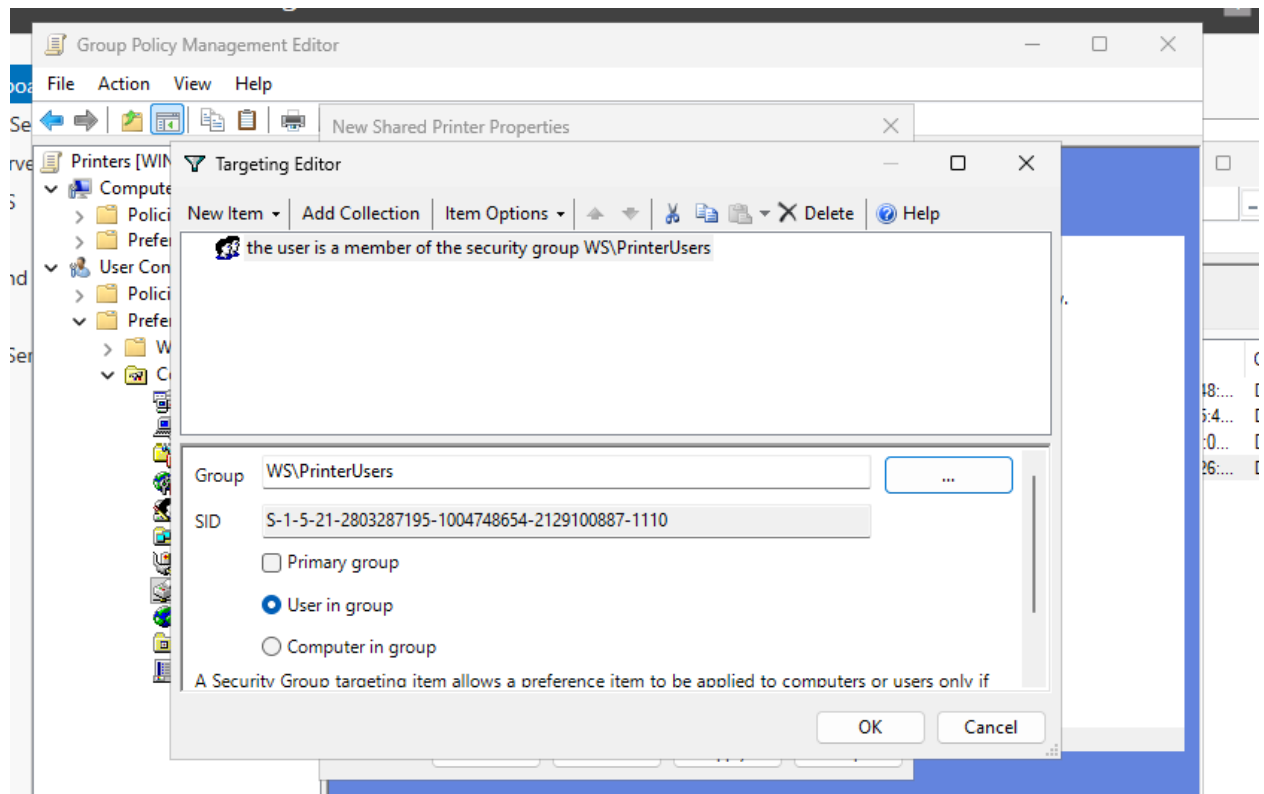
In the opened window in the Shared path click on three dots ... This action will open an AD Printer list, and because we enable the printer to be listed in the directory, it will show up there.



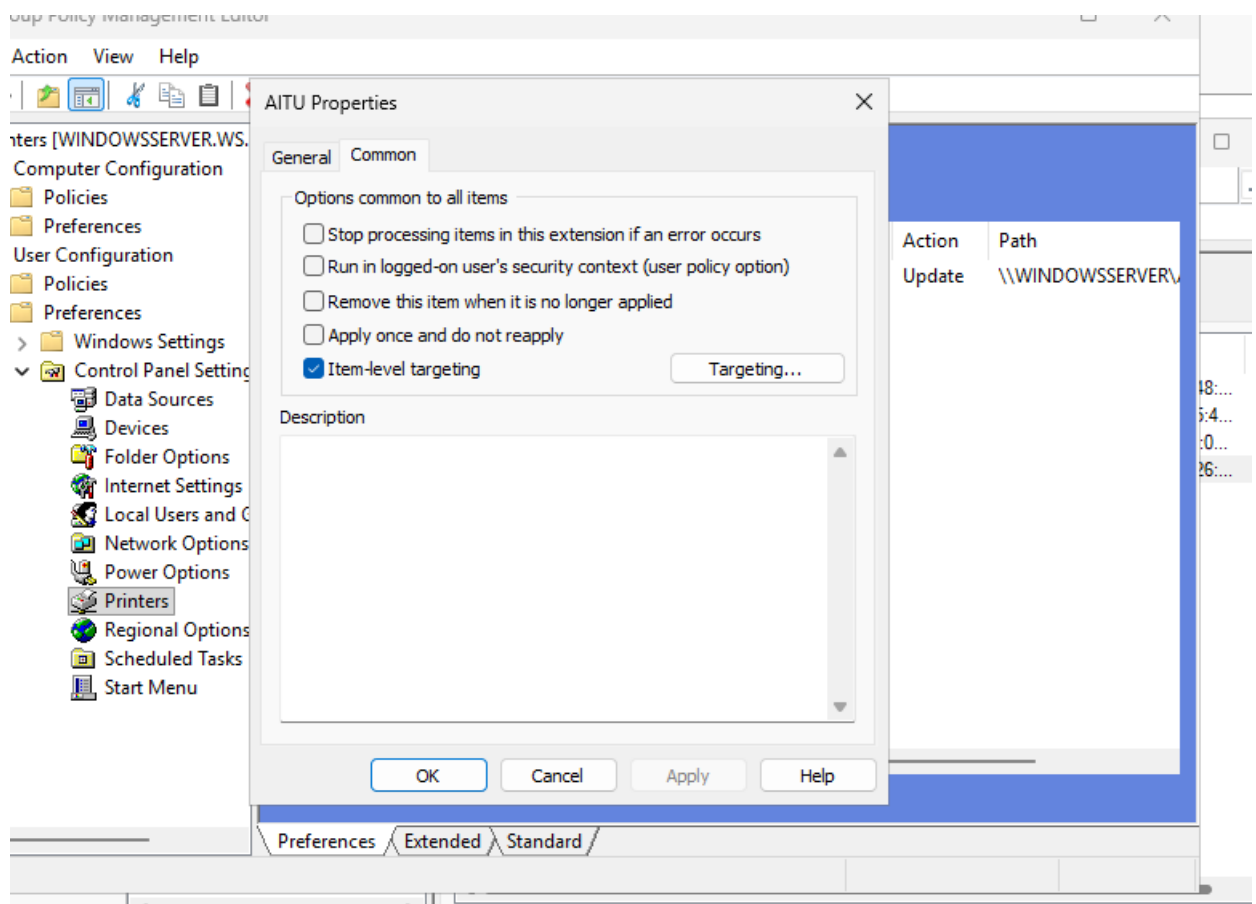
Action type must be Update. Go to the Common tab, select Item level targeting. Press Targeting.

In the opened window select New item -> Security group-> Press button ... -> and type there the name of the security group that you created in step 18. Press Ok.

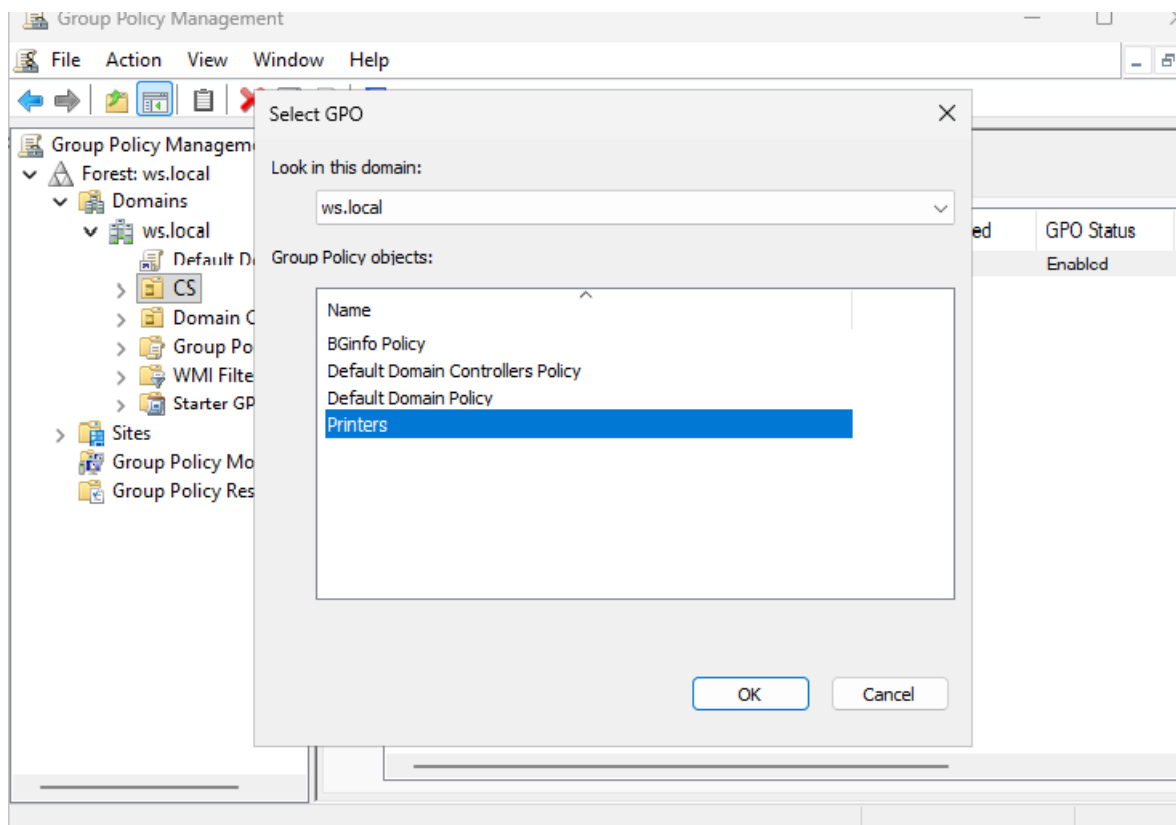




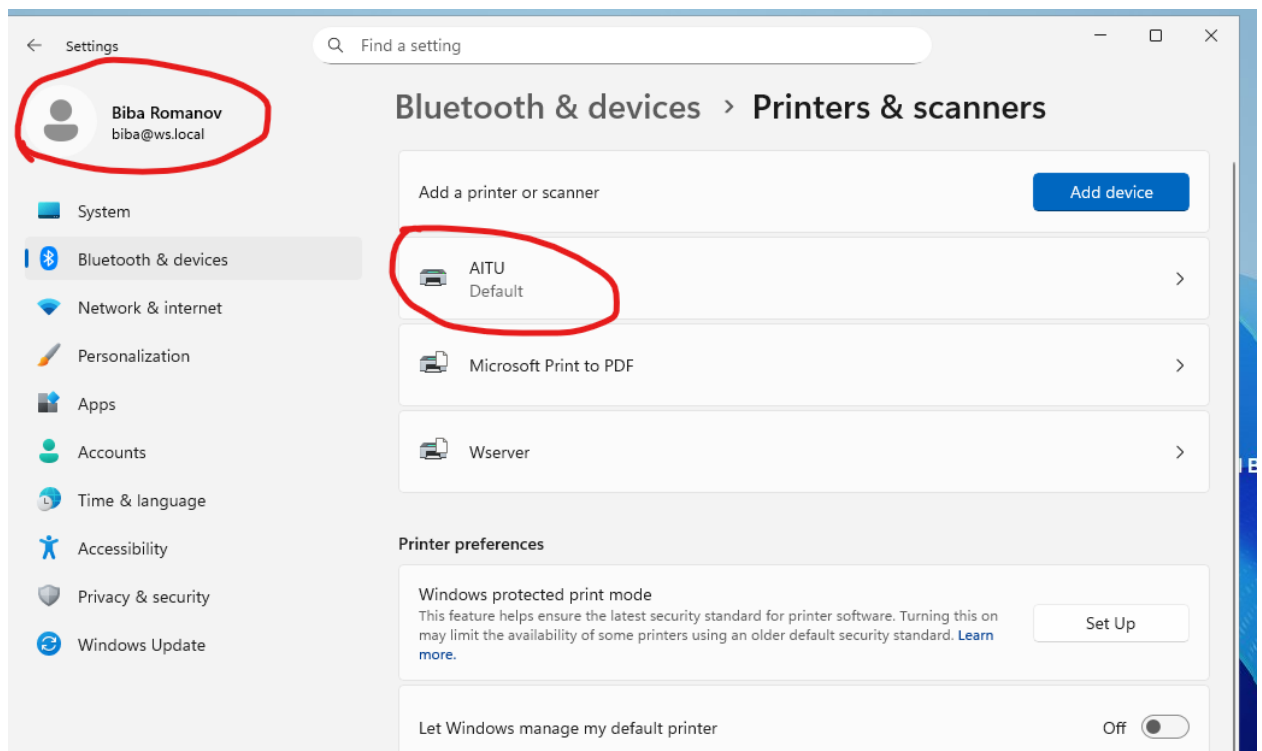
On the previous step we targeted the shared printer to anyone that is within a special security group. Press Apply -> Ok.



Link your main OU (in my case it was CS) to the existing GPO, and select the Printers GPO



Sign out from the admin's account and sign in to any user in your network. Go to Printers & scanners settings. You can see the installed printer in the list.

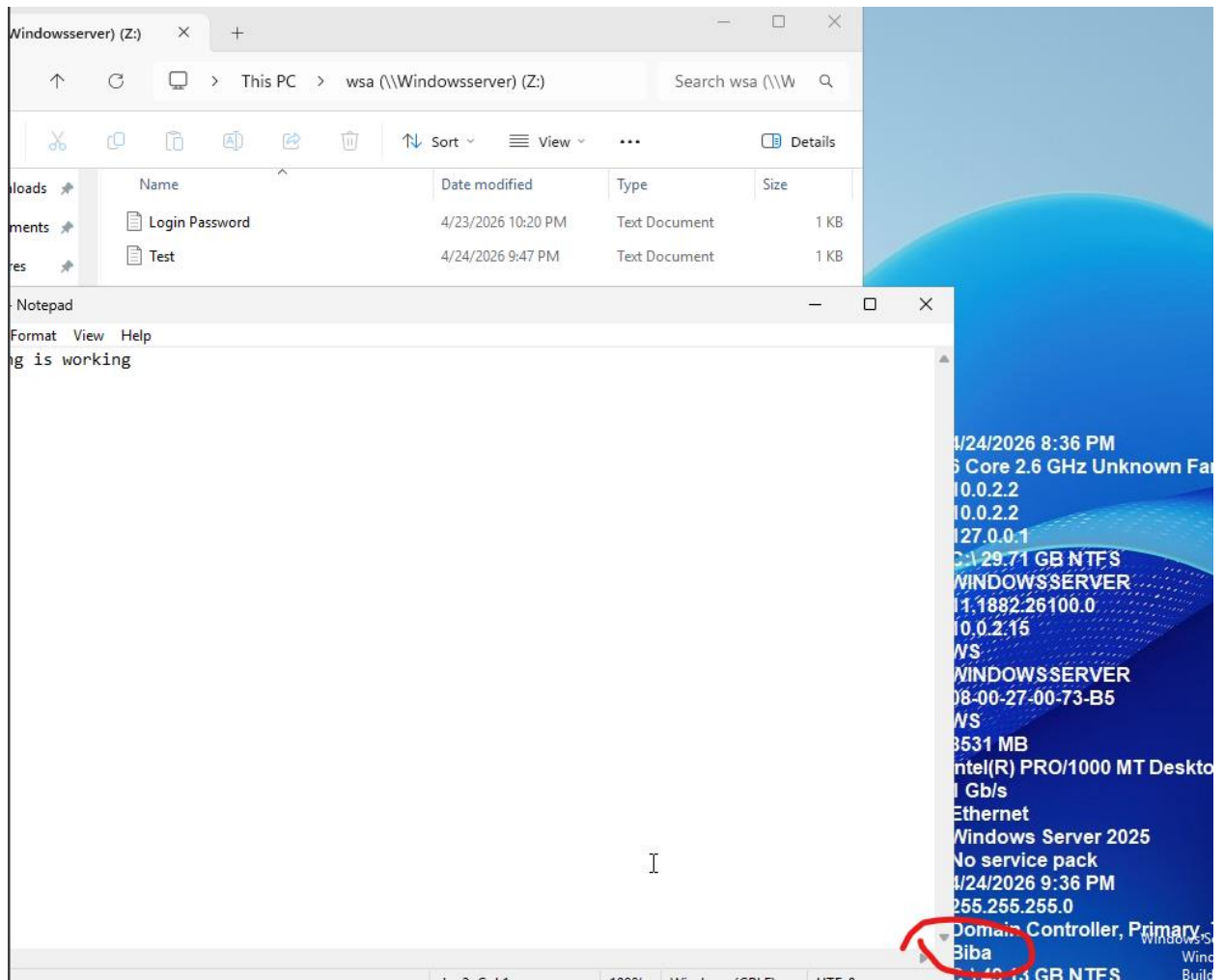


Part 5.

Tasks: 1.

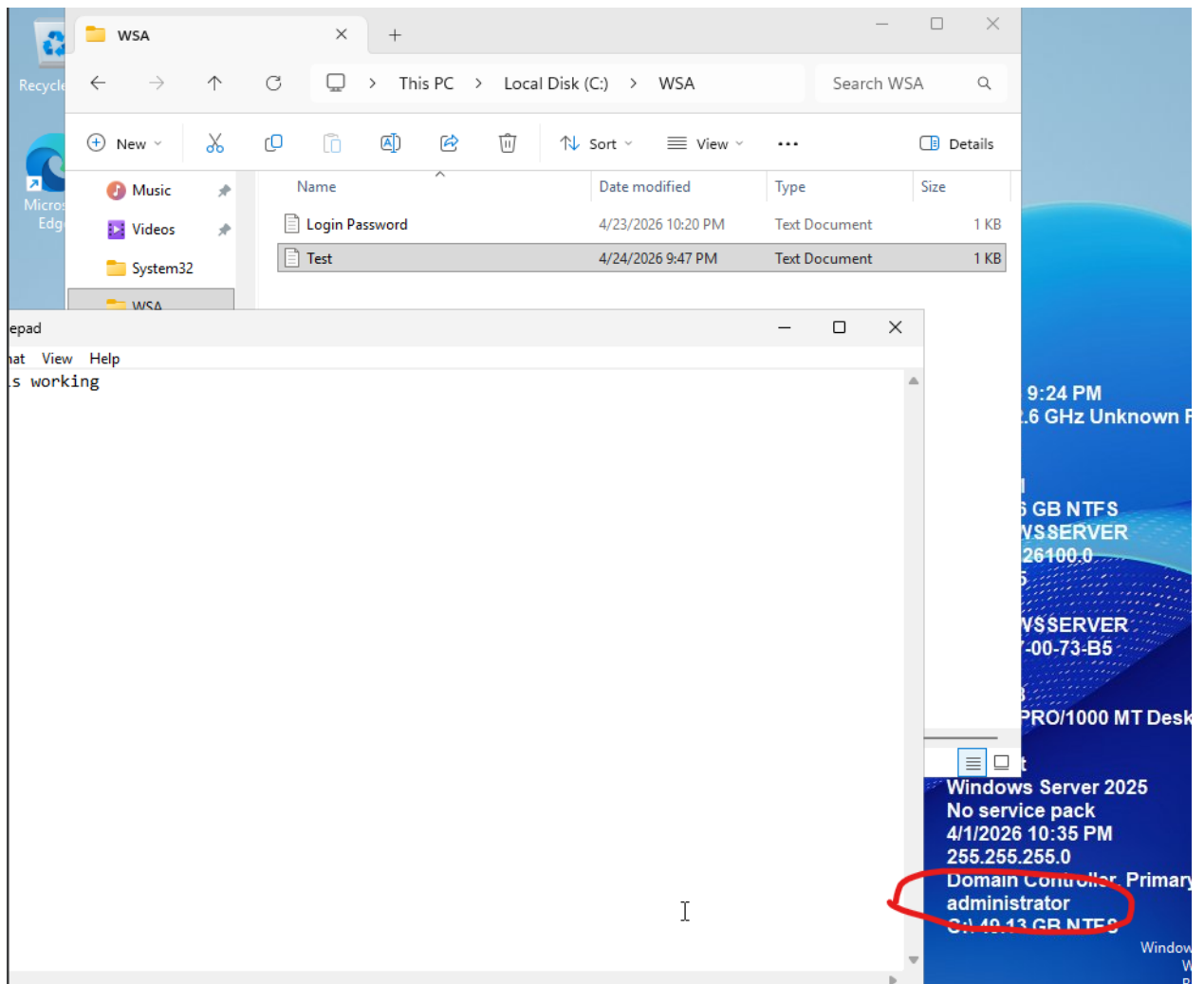
For Part 1:

a) Let some of your users modify the network drive that you previously created.



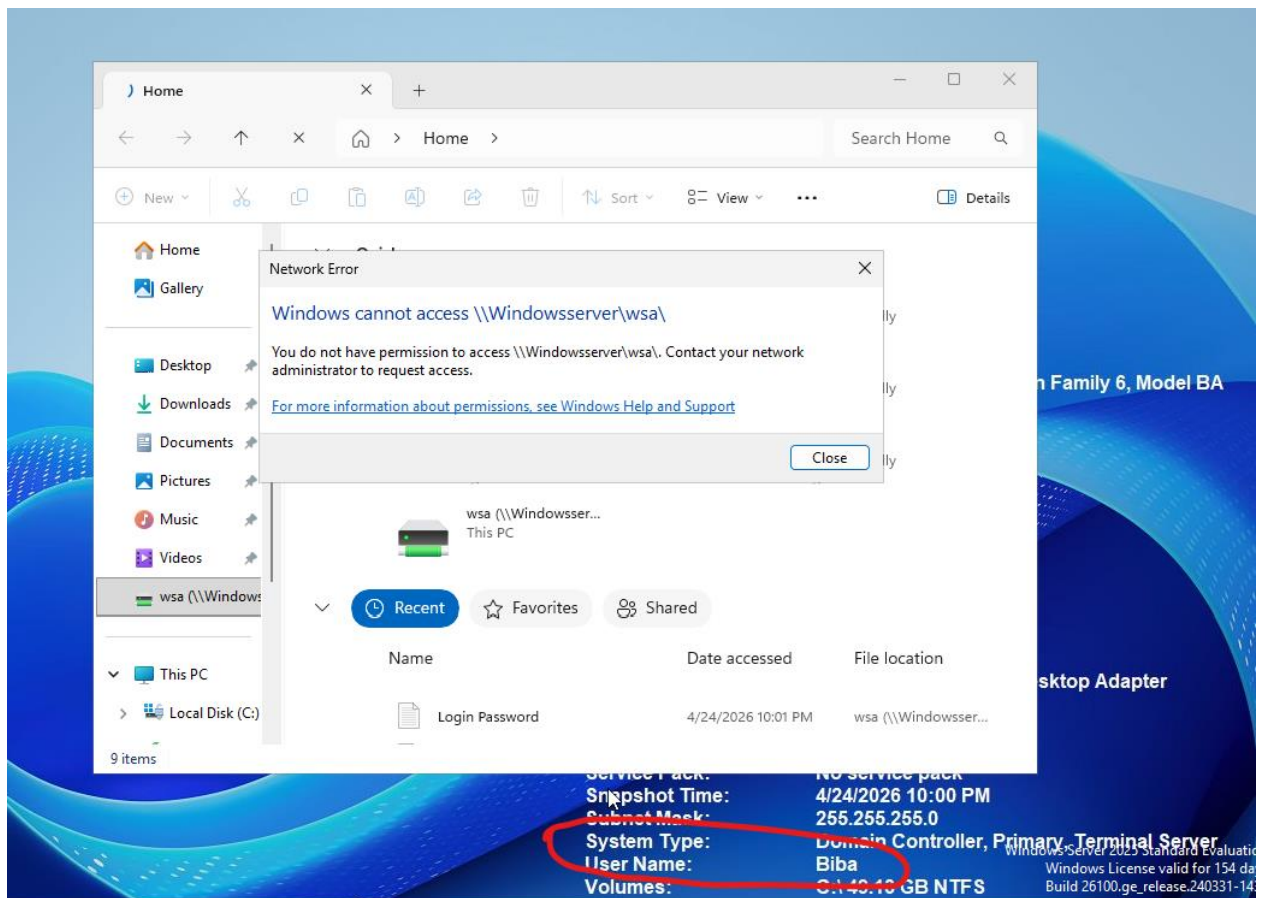
User created and modified a file in the shared network folder to test write permissions.

b) Create through user some folders and files and check whether these changes were visible through admin's account.



Administrator accessed the shared folder and verified that the file created and modified by the user is visible and unchanged, confirming that shared network storage is functioning correctly

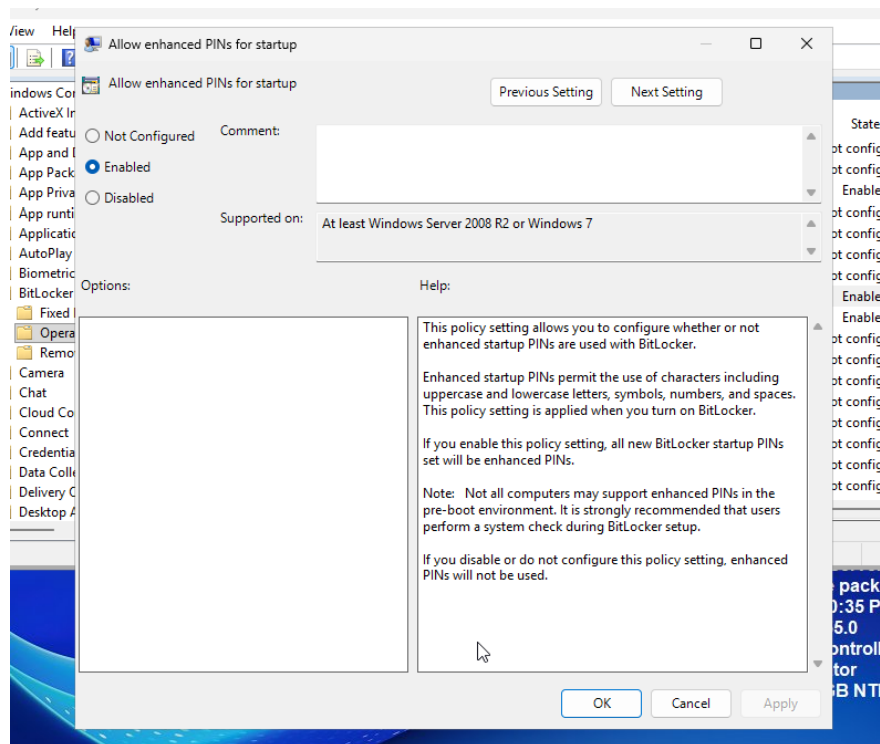
c) Restrict access for the other user from the network drive and check the results of the restrictions.



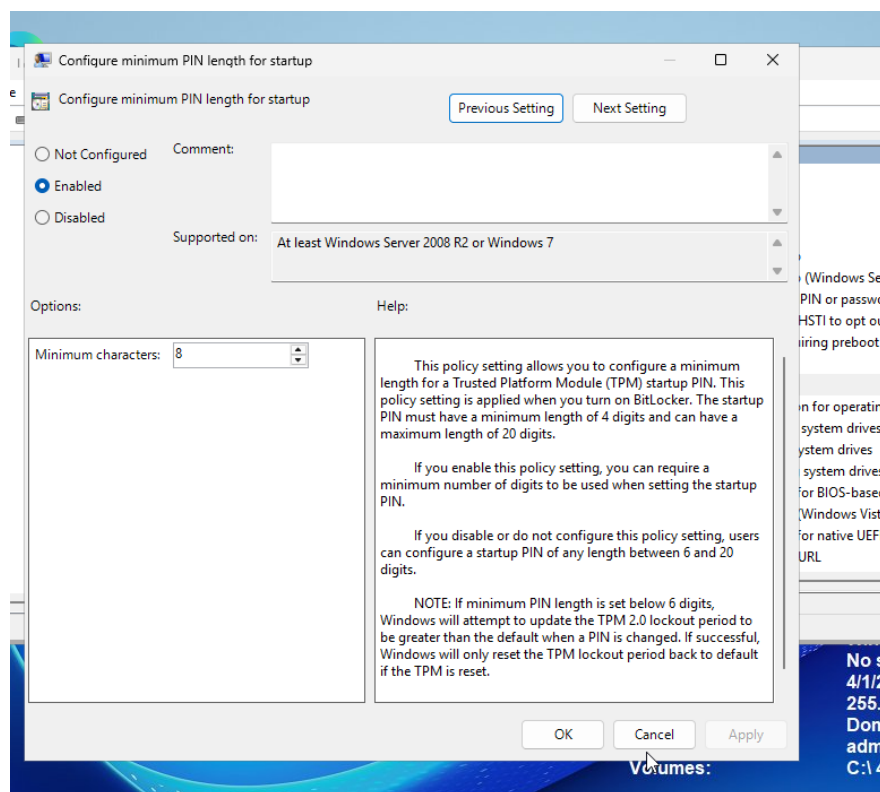
Access to the shared folder was restricted for a specific user. When the user attempted to open the folder, access was denied, confirming that permissions are enforced correctly.

2. For Part 2: a) Set your own requirements for PIN password.

I enabled enhanced PINs for startup

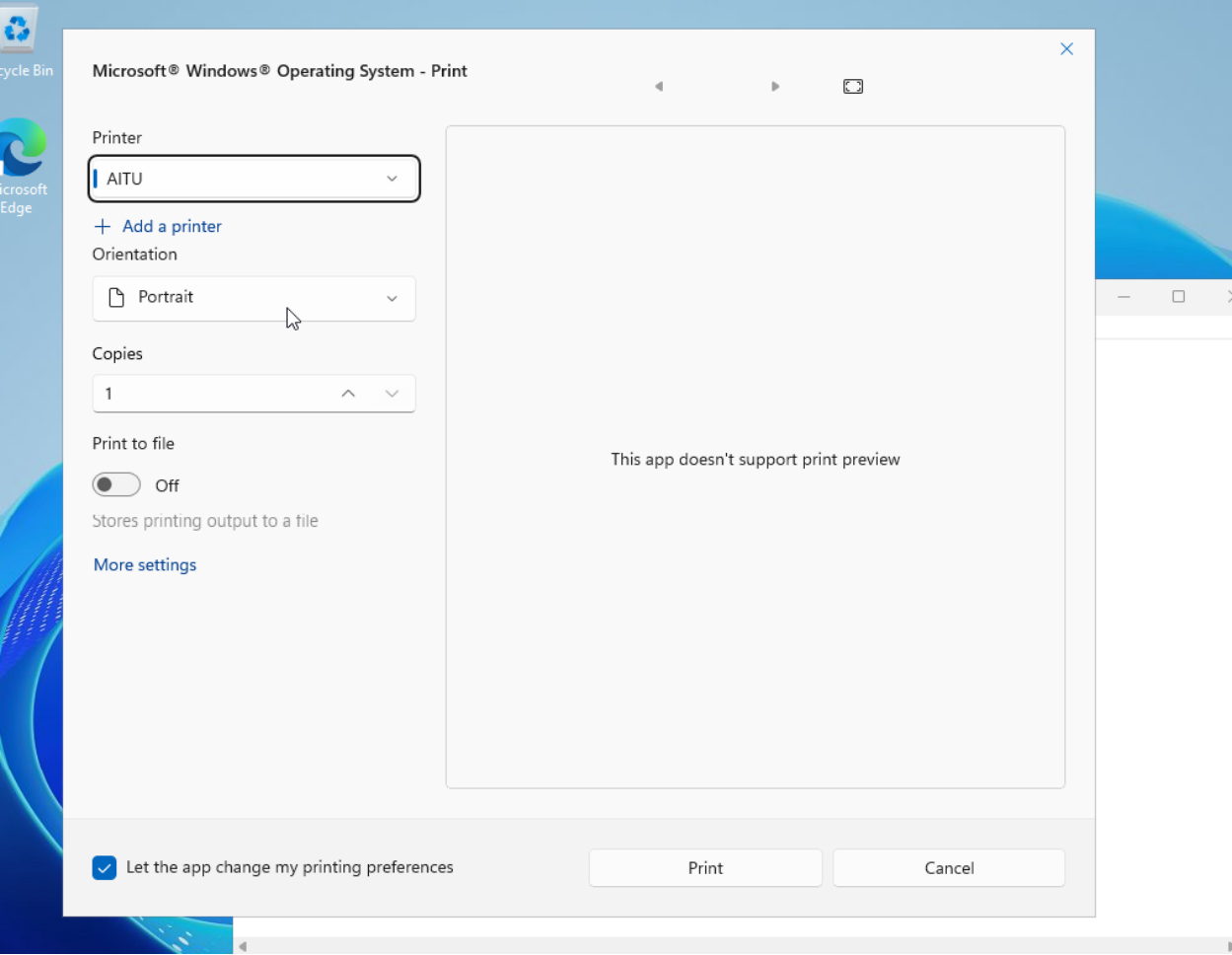


I set min length to 8 characters

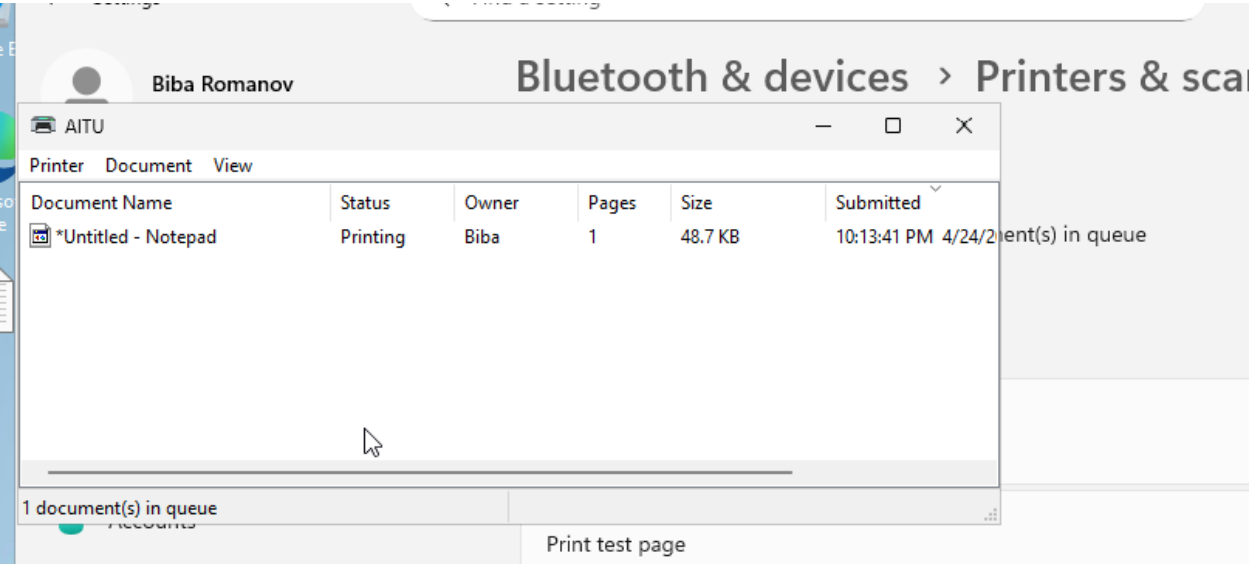


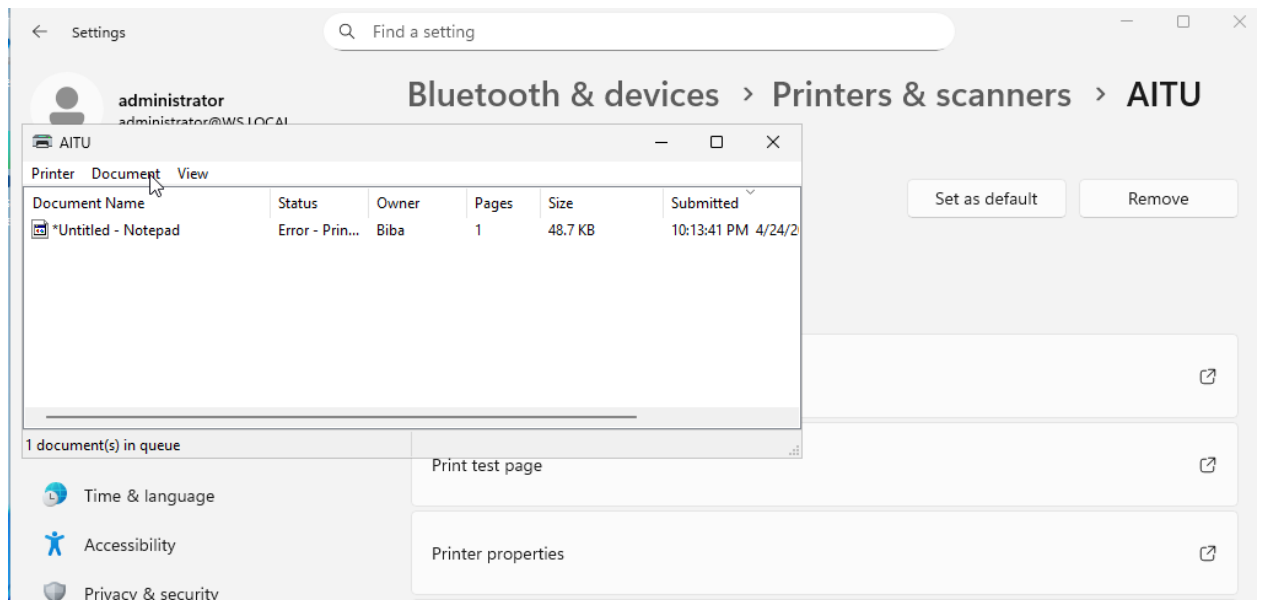
I configured BitLocker to allow enhanced PINs and set a minimum PIN length to increase security during system startup.

3. For Part 4: check how the printer is working now. In the user's account, create any text file, and print it using the created printer. Check in Printers & scanners window that 1 document is in queue now. Then sign in to the administrator's account and check in the Print Management's Printers list whether your printer has some changes in Queue status and Jobs in Queue or not. Show the Printer Queue.



My document is in the queue from Biba side:





Administrator verified the print job in Print Management. The queue shows one document, confirming that printer deployment and communication are functioning correctly.

THE END!